

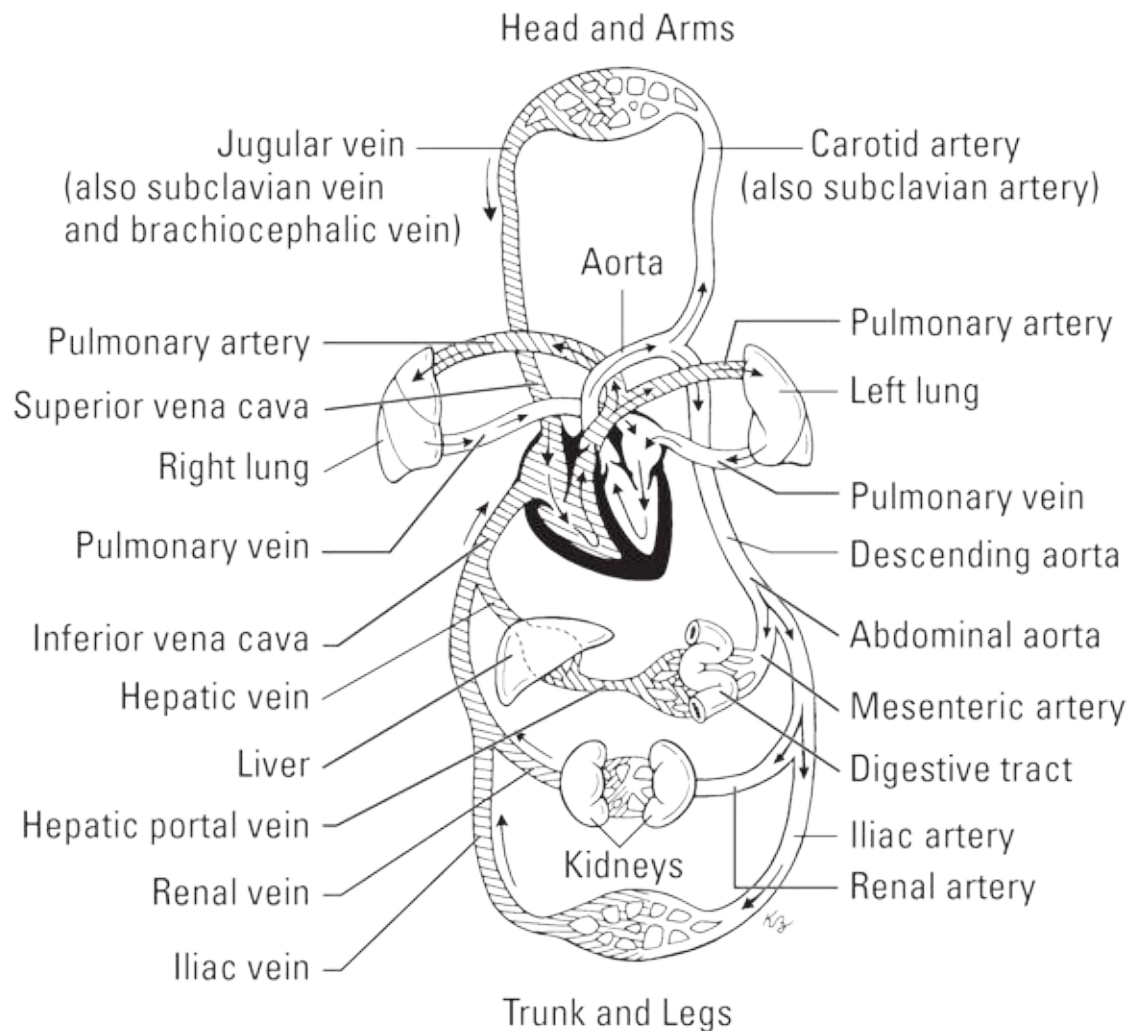


Illustration by Kathryn Born, MA

FIGURE 11-1: The cardiovascular system is a closed double loop.

The heart is separated into right and left halves by the septum, allowing blood to flow through each half without mixing. With each *cardiac cycle* (a single set of muscle contractions), blood moves through the right and left sides simultaneously, thus creating the two circuits — the right side destined for the lungs and the left to, well, everywhere else.

These two circuits have distinct functions:

» **Pulmonary:** The *pulmonary circuit* carries blood to and from the lungs for gas exchange. Deoxygenated blood saturated with carbon dioxide enters the right side of the heart from the *inferior* and *superior vena cavae*. From there, it is pumped out through the *pulmonary trunk* to *pulmonary arteries* to capillaries in the lungs, where the carbon dioxide

departs the system. That same blood, freshly loaded with oxygen (*oxygenated*), then returns to the left side of the heart through the *pulmonary veins*, where it enters the second circuit.



WARNING Though we tend to associate arteries with oxygenated blood and veins with deoxygenated blood, the terms refer to their structure. Arteries are adapted to carry blood away from the heart under high pressure, while veins are the opposite. As such, the pulmonary arteries carry deoxygenated blood and the pulmonary veins carry oxygenated blood.

- » **Systemic:** The *systemic circuit* uses the oxygenated blood to maintain a constant internal pressure and flow rate around the body's tissues, allowing them access to the oxygen and other vital nutrients in the blood. The oxygenated blood enters the left side of the heart through the *pulmonary veins* and is pumped out through the *aorta* to be distributed throughout the body. Branching off the aorta is the *coronary artery*, creating a side circuit to provide resources to the *myocardium* (the muscular wall of the heart). From here, the *cardiac veins* carry the deoxygenated blood to the right side. Blood from the rest of the body is brought back to the heart in the vena cavae.

Although cutely depicted in popular culture as uniformly curvaceous, the heart actually looks more like a blunt, muscular cone resting on the diaphragm. The sternum (breastbone) and third to sixth costal cartilages of the ribs provide protection in front of (ventrally to) the heart. Behind it lie the fifth to eighth thoracic vertebrae. (Check out [Chapter 7](#) to review the skeleton.) Two-thirds of the heart lies to the left of the body's center nestled into the left lung.

While in the body, the heart is not positioned as you see in diagrams. It is twisted slightly so the right side is anterior (more forward) to the left side. The *apex* is the point of the heart and the *base* is the opposite end where all the blood vessels attach.



TIP

So, the base is the top and the apex is the bottom? Isn't that

backwards? Well, yes, sort of. But as you'll see by the time you finish this chapter, contraction starts at the apex to push the blood toward the base. If you think of it that way, you won't get the two confused.

Three layers make up the wall of the heart:

- » **Epicardium:** On the outside lies the *epicardium*, which is composed of connective tissue containing adipose tissue (fat) to protect the underlying muscle. It forms a framework for nerves as well as the blood vessels that supply nutrients to the heart's tissues.
- » **Myocardium:** Beneath the epicardium lies the *myocardium*, which is composed of organized layers and bundles of cardiac muscle tissue.
- » **Endocardium:** The *endocardium*, the heart's interior lining, is composed of simple squamous endothelial cells. This layer is continuous with all of the systems' blood vessels.



TIP

Too much to remember? To keep the layers straight, turn to the Greek roots. *Epi-* is the Greek term for “upon” or “on” whereas *endo-* comes from the Greek *endon* meaning “within.” The medical definition of *myo-* is “muscle.” And *peri-* comes from the Greek term for “around” or “surround.” Hence the *epicardium* is *on* the heart, the *endocardium* is *inside* the heart, the *myocardium* is the *muscle* between the two, and the *pericardium* surrounds the whole package. By the way, the root *cardi-* comes from the Greek word for heart, *kardia*.

Since the heart beats with such gusto it's isolated from all of the other structures in the mediastinum in a protective sac called the *pericardium*. The pericardium is composed of multiple layers of connective tissue. The outermost layers comprise the *parietal pericardium*, and the innermost is the *visceral pericardium* (which is also known as the epicardium). Space is left between these layers creating the *pericardial cavity*. This space, albeit tiny, is filled with a serous fluid to reduce friction, preventing damage to the heart's walls as they are moved with each set of contractions.

Have you got the beat? Let's try a few practice questions:

1-5 Match the description to its anatomical term.

- a. Pericardium
- b. Pulmonary circuit
- c. Systemic circuit
- d. Epicardium
- e. Septum

- 1. _____ The system for gaseous exchange in the lungs
- 2. _____ The system for maintaining a constant internal environment in other tissues
- 3. _____ The membranous sac that surrounds the heart
- 4. _____ The wall that divides the heart into two cavities
- 5. _____ The layer providing the heart's wall with nutrients

6 What's the difference between the pulmonary and systemic circuits?

- a. The pulmonary circuit supplies blood to the brain, and the systemic circuit supplies it everywhere else.
- b. The pulmonary circuit supplies blood to the extremities, and the systemic circuit supplies blood to the internal organs.
- c. The pulmonary circuit supplies blood to the skin, and the systemic circuit supplies blood to the underlying muscles.
- d. The pulmonary circuit supplies blood to the left side of the body, and the systemic circuit supplies blood to the right side.
- e. The pulmonary circuit travels to and from the lungs, and the systemic circuit keeps pressure and flow up for the body's tissues.

7 True or False: The heart is centrally located in the chest.

8 Which blood vessel contains oxygenated blood?

- a. pulmonary artery
- b. pulmonary vein

- c. pulmonary trunk
- d. inferior vena cava
- e. superior vena cava

9 Only one of the heart wall's three layers contains muscle tissue. Which one?

- a. Pericardium
- b. Epicardium
- c. Myocardium
- d. Endocardium
- e. Endothelium

Finding the Key to the Heart's Chambers

Each cavity of the heart is divided into two chambers: the right and left atria on top and the right and left ventricles on bottom. The ventricles are quite a bit larger than the two atria up top. Yet the proper anatomical terms for their positions refer to the atria as being “superior” (above) and the ventricles “inferior” (below). In this case, size isn't the issue at all. Between each atrium and ventricle is an *atrioventricular valve (AV valve)*, which have leaflets, or *cusps*, that open and close to control entry into the ventricles. The exits of the ventricles (into the arteries) are guarded by *semilunar valves*. The three cups that form these valves are half-moon shaped, hence the name.

The atria

The atria are sometimes referred to as “receiving chambers” because they receive blood returning to the heart through the veins. Each atrium has two parts: a *principal cavity* with a smooth interior surface and an *auricle* (atrial appendage), a smaller, dog-ear-shaped pouch with muscular ridges inside called *pectinate muscles*. The atria have thin walls as they only need to generate enough force to pump blood into the ventricles. The right atrium is slightly larger than the left and the *interatrial septum* lies between them.

Deoxygenated blood passively (pressure is lower in the atrium than in the blood vessels) fills the right atrium through three openings:

- » The *superior vena cava*, which has no valve and returns blood from the head, thorax, and upper extremities.
- » The *inferior vena cava*, which returns blood from the trunk and lower extremities.
- » The *coronary sinus*, returns blood from the heart, and is covered by the ineffective *Thebesian valve*.

At the bottom of the right atrium is the tricuspid valve. Its three flaps open when the pressure in the right ventricle drops below that of the right atrium. This allows the blood to flow into the right ventricle. Attached to the tricuspid valve on the ventricular side (underneath) are strong strings called *chordae tendineae*. These fuse with *papillary muscles* in the walls of the ventricle. The chordae tendineae are pulled tight when the ventricle contracts to hold the tricuspid valve closed, preventing the backflow of blood.

The left atrium's principal cavity contains openings for the four *pulmonary veins*, two from each lung, which have no valves. Frequently, the two left veins share a common opening. The left auricle, a dog-ear-shaped blind pouch, is longer, narrower, and more curved than the right. The left atrium's AV opening is smaller than on the right and is guarded by the *mitral*, or *bicuspid*, valve. Though it has only two cusps, it contains chordae tendineae and functions the same as the tricuspid valve.

The ventricles

The heart's ventricles are sometimes called the pumping chambers because it's their job to receive blood from the atria and pump it out through the arteries. More force is needed to move the blood great distances, so the myocardium of the ventricles is thicker than that of either atrium, and the myocardium of the left ventricle is thicker than that of the right. (It takes greater pressure to maintain the systemic circuit.) The two ventricles are separated by a thick wall of muscle called the *interventricular septum*.

The right ventricle only has to move blood to the lungs, so its myocardium is only one-third as thick as that of its neighbor to the left. Roughly triangular in shape, the right ventricle occupies much of the *sternocostal* (front) surface of

the heart and joins the *pulmonary trunk* just left of the AV opening. The right ventricle extends downward toward where the heart rests against the diaphragm. Cardiac muscle in the ventricle's wall is in an irregular pattern of bundles and bands called the *trabeculae carneae*. The circular opening into the pulmonary trunk is covered by the *pulmonary valve* that opens when the ventricle contracts (though not immediately). When the ventricle relaxes, the blood from the pulmonary artery tends to flow back toward the ventricle, filling the pockets of the cups and causing the valve to close. Openings into the ventricles are surrounded by strong, fibrous rings sometimes referred to as the *cardiac skeleton* (though there are definitely no bones here).

Longer and more conical in shape, the left ventricle's tip forms the apex of the heart. More ridges are packed more densely in the muscular trabeculae carneae. Its opening to the aorta is protected by the *aortic valve* that is larger, thicker, and stronger than the pulmonary valve's cups. The blood enters the ascending aorta into the aortic arch where three vessels branch off:

- » The brachiocephalic trunk that will divide into the right common carotid artery and the right subclavian artery
- » The left common carotid artery
- » The left subclavian artery

Pump up your practice time with these questions related to the chambers of the heart:

10-25 Use the terms that follow to identify the heart's major vessels shown in [Figure 11-2](#).

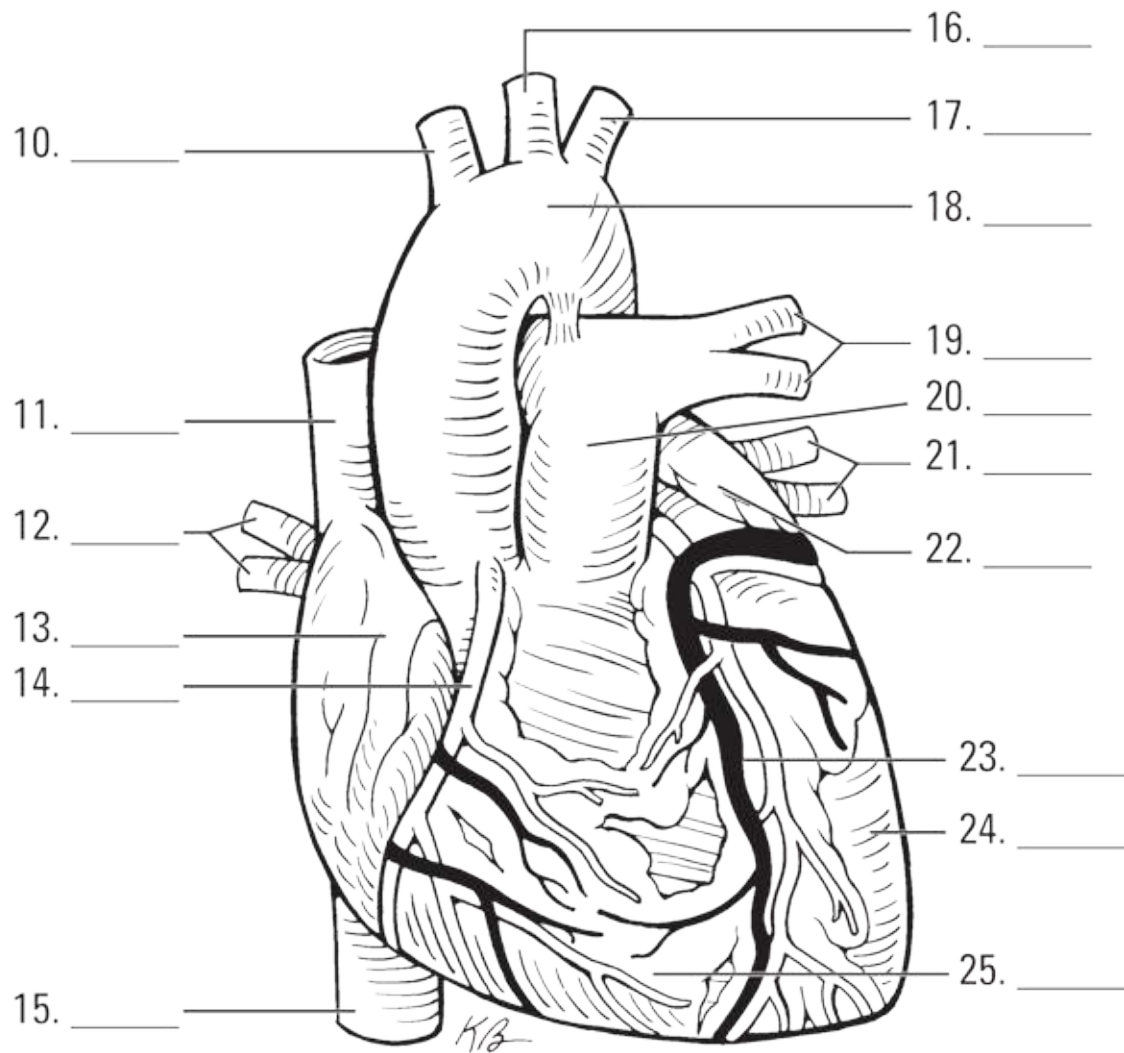


Illustration by Kathryn Born, MA

FIGURE 11-2: The heart and major vessels.

- a. Left pulmonary veins
- b. Left ventricle
- c. Brachiocephalic trunk
- d. Right pulmonary veins
- e. Right ventricle
- f. Left subclavian artery
- g. Right coronary artery
- h. Aortic arch

- i. Great (left) cardiac vein
- j. Superior vena cava
- k. Left common carotid artery
- l. Left pulmonary arteries
- m. Left atrium
- n. Inferior vena cava
- o. Pulmonary trunk
- p. Right atrium

26-29 Use the terms that follow to identify the heart valves shown in [Figure 11-3](#).

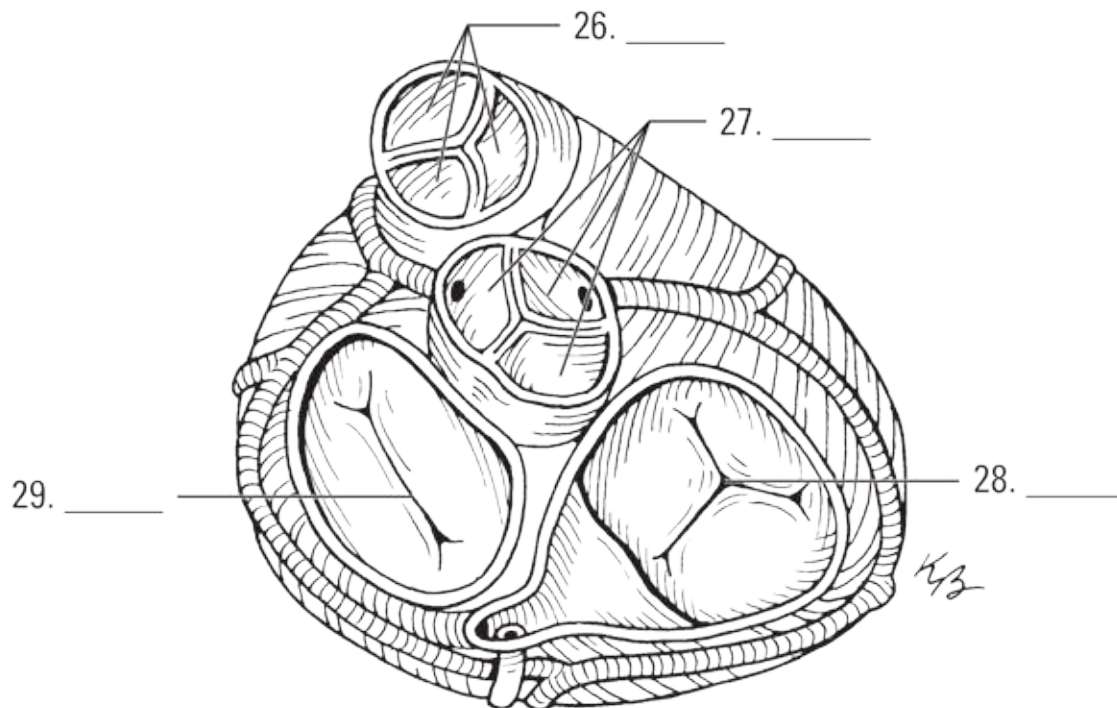


Illustration by Kathryn Born, MA

FIGURE 11-3: The heart valves.

- a. Tricuspid valve
- b. Pulmonary valve
- c. Aortic valve
- d. Mitral valve

30 Why is the myocardium of the ventricles thicker than the myocardium of the atria?

- a. The myocardium keeps valves in place, and the ventricular valves have to hold back greater blood pressure with each contraction.
- b. The myocardium is the inner lining of the heart, and the ventricles must hold greater volumes of blood between contractions.
- c. The myocardium is the muscular tissue of the heart, and more force is needed in chambers pumping blood greater distances.
- d. The myocardium consists mostly of connective tissue, and the ventricles are more intricate than the atria.

31 The superior vena cava enters the heart by way of the

- a. left ventricle.
- b. right ventricle.
- c. left atrium.
- d. right atrium.

32 What holds the cusps of the atrioventricular valves in place?

- a. Supporting ligaments
- b. The chordae tendineae
- c. The trabeculae carneae
- d. Pectinate muscles
- e. The papillary muscles

33 What covers the atrioventricular opening between the right atrium and the right ventricle?

- a. Bicuspid valve
- b. Tricuspid valve
- c. Pulmonary valve

- d. Mitral valve
- e. Aortic valve

34-40 Match the following descriptions with the proper anatomical terms.

- a. Tricuspid valve
 - b. Mitral valve
 - c. Superior vena cava
 - d. Pulmonary valve
 - e. Inferior vena cava
 - f. Aortic valve
 - g. Pulmonary vein
34. _____ Returns blood to the heart from the head, thorax, and upper extremities
35. _____ Valve located between the right atrium and right ventricle
36. _____ Valve located between the right ventricle and pulmonary trunk
37. _____ Returns blood to the heart from the trunk and lower extremities
38. _____ Valve located between the left atrium and left ventricle
39. _____ Semilunar valve located in the left ventricle
40. _____ Returns blood from the lungs

Conducting the Heart's Music

The mighty, nonstop heart keeps up its rhythm because of a carefully choreographed dance of electrical impulses called the *cardiac conduction system* that has the power to produce a spontaneous rhythm and conduct an electrical impulse. The feature unique to cardiac muscle, the presence of *intercalated discs*, plays an essential role here. These undulating double membranes separate adjacent cardiac muscle fibers with *gap junctions* permitting ions to pass between the cells. This allows the spread of the *action potential* of the electrical impulse throughout the myocardium, synchronizing cardiac muscle contractions.

The orchestra

Five structures play key roles in this dance — the *sinoatrial node*, *atrioventricular node*, *atrioventricular bundle (bundle of His)*, *bundle branches*, and *Purkinje fibers*. (You can see them in [Figure 11-4](#).) Each is formed of highly tuned, modified cardiac muscle. Rather than both contracting and conducting impulses as other cardiac muscle does, these structures specialize in conduction alone, setting the pace for the rest of the heart. Following is a bit more information about each one:

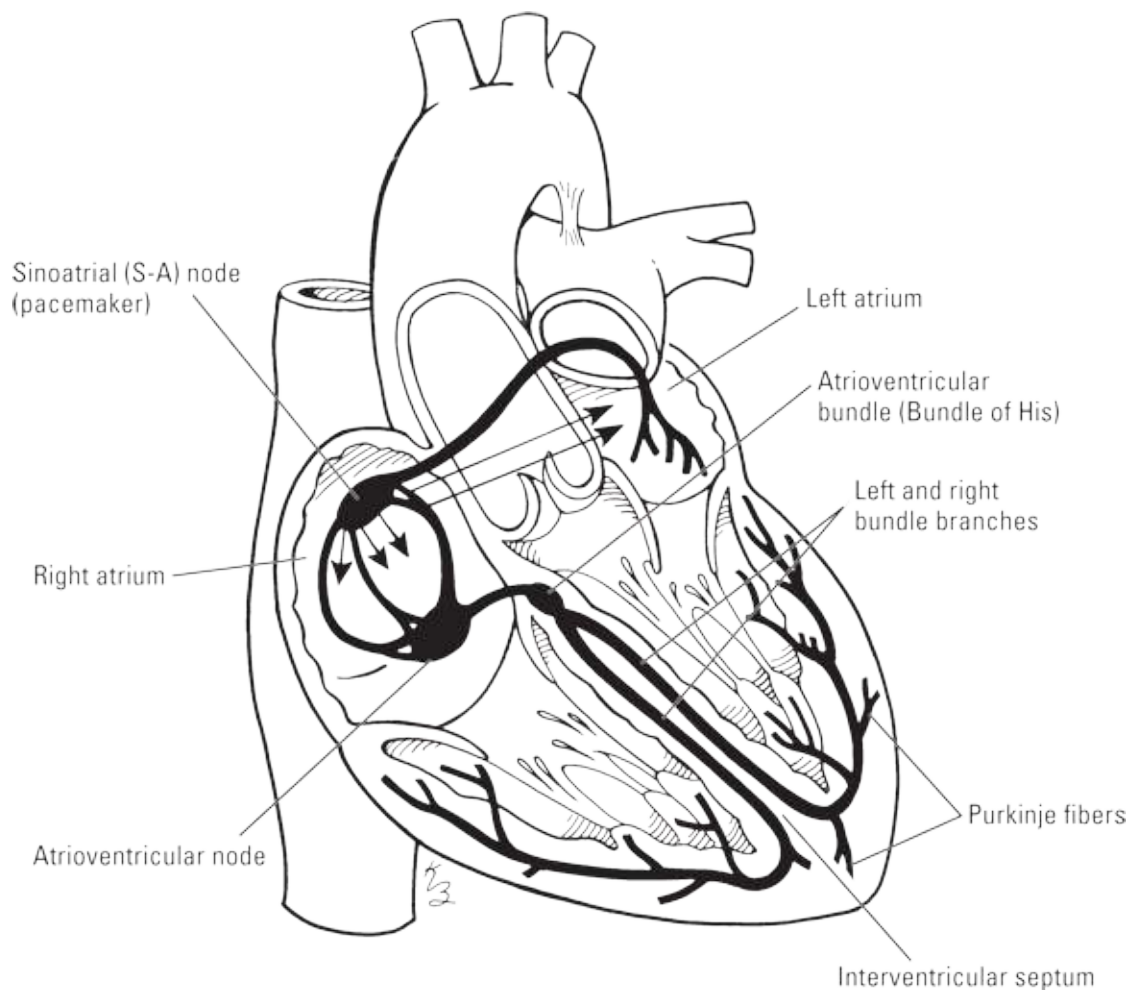


Illustration by Kathryn Born, MA

FIGURE 11-4: The conductive system of the heart.



» REMEMBER **Sinoatrial node (SA node):** This node really is the pacemaker of

the heart. Located at the junction of the superior vena cava and the right atrium, this small knot, or mass, of specialized cardiac muscle cells initiates an electrical impulse that moves over the musculature of both atria. This network of conducting fibers forms the *atrial syncytium*, causing atrial walls to contract simultaneously and emptying blood into both ventricles.

- » **Atrioventricular node (AV node):** The impulse that starts in the SA node moves to this mass of conducting fibers that's located at the junction of all four chambers of the heart. Fibers carrying impulse from the SA node to the AV node are smaller, slowing it down to ensure that the ventricles contract after the atria.
- » **Atrioventricular bundle:** From the AV node, the impulse moves into the atrioventricular bundle, also known as the *AV bundle* or *bundle of His* (pronounced "hiss").
- » **Bundle branches:** The atrioventricular bundle breaks into two branches that extend down the sides of the interventricular septum under the endocardium to the heart's apex.
- » **Purkinje fibers:** At the apex, the bundles branch into Purkinje fibers, which spread throughout the myocardium forming the *ventricular syncytium*. Both ventricles then contract together in an upward, twisting motion beginning at the apex and moving toward the base of the heart, forcing blood into the aorta and pulmonary trunk.

The music

Though we can stimulate the SA node through the autonomic nervous system, it is self-stimulating, so we need only do so to alter the normal, or *sinus rhythm*. While we tend to think of contraction as a quick event, it's more like a phase. When the chambers are contracting, they are said to be in *systole* (pronounced *sis-toe-lee*); when relaxed they are in *diastole* (pronounced *di-as-toe-lee*). To begin the cardiac cycle, all chambers are in systole and all valves are closed. It proceeds as follows:

1. Blood trickles into the atria from the veins, increasing the internal pressure.
2. When the pressure in the atria rises above that of the ventricles (they

would be relaxing at this time thus dropping the pressure) the AV valves are pushed open.

3. Blood passively fills the ventricles until they are about 70-percent full. At this point, the pressures of the chambers have equalized.
4. Atrial systole begins, pushing the remaining 30 percent of the blood into the ventricles.
5. The atria begin to relax as the ventricles now begin systole. This drops the pressure of the atrium creating a vacuum. The cusps of the AV valves are thus pulled upwards. However, the papillary muscles are contracting, pulling on the chordae tendineae, preventing the cusps from being pulled into the atrium.
6. At this point, all valves are closed, the atria are in diastole, and the ventricles are in systole.
7. The ventricles continue contracting to increase the internal pressure. Once it rises above the pressure inside the arteries, the semilunar valves are pushed open.
8. Blood enters the arteries to be carried elsewhere and the ventricles begin diastole.
9. Again, this creates a vacuum pulling the blood backwards, this time toward the ventricles.
10. Blood catches in the cups of the semilunar valves, sealing them shut (there are no chordae tendineae here). And now we're back to the beginning.



WARNING It's easy to fall into the trap of thinking that contractions open and close the valves. However, this is not so. With the exception of the AV valves closing (which is initiated by a pressure change but maintained by ventricular contraction), all valve movements are triggered by pressure changes.

The performance

The cardiac cycle generates the forces necessary to propel the blood along the

two circuits. The spread of impulse is measurable through an electrocardiogram (EKG or ECG) which generates the familiar spikes visible on a heart rate monitor. Keep in mind, though, that these spikes precede the contractions and are not the contractions themselves.

Similarly, when listening to the heartbeat, the sounds you hear are not the contractions. A healthy heart makes a “lub-dub” sound as it beats because a single cardiac cycle contains two separate contractions: The first sound (the “lub”) is heard most clearly near the apex of the heart and comes at the beginning of ventricular systole — the sound you hear is the atrioventricular valves snapping shut. It’s lower in pitch and longer in duration than the second sound (the “dub”), heard most clearly over the second rib, which results from the semilunar valves closing during ventricular diastole.

Give these practice questions a try to see if you’ve got the rhythm of all of this:



EXAMPLE Q. Cardiac tissue is distinctive microscopically because of the presence of

- a. hemoglobin.
- b. intercalated discs.
- c. fibrin.
- d. myofibrils
- e. ganglia.

A. The correct answer is intercalated discs. These undulating double membranes separate adjacent cardiac muscle fibers, so they’re unique to cardiac tissue.

41 Why is the sinoatrial node (SA node) called the heart’s “pacemaker”?

- a. It interrupts irregular heart rhythms to set a steadier pace.
- b. It initiates an electrical impulse that causes atrial walls to contract simultaneously.

- c. It speeds up and slows down the heart, depending on the body's activities.
- d. It tells the ventricles to contract simultaneously.
- e. It supplies a boost of energy to jump-start the heart when it slows.

42 True or False: The chordae tendineae open the AV valves.

43 Choose the correct conductive pattern.

- a. S-A node → bundle of His → bundle branches → A-V node → Purkinje fibers
- b. A-V node → S-A node → Purkinje fibers → bundle of His → bundle branches
- c. S-A node → A-V node → bundle of His → bundle branches → Purkinje fibers
- d. S-A node → Purkinje fibers → bundle of His → bundle branches → A-V node

44 Which valve action results from the pressure in the ventricle being lower than that of the atria?

- a. semilunar valves open
- b. semilunar valves close
- c. AV valves open
- d. AV valves close

45 Draw brackets on the EKG in [Figure 11-5](#) to show atrial and ventricular systole.

46 True or False: The atrial syncytium is comprised of the conducting fibers of the atrium that ensure the atria contract before the ventricles.

47 Starting with a drop of blood returning to the heart from the legs, trace its path until its eventual return to the legs.

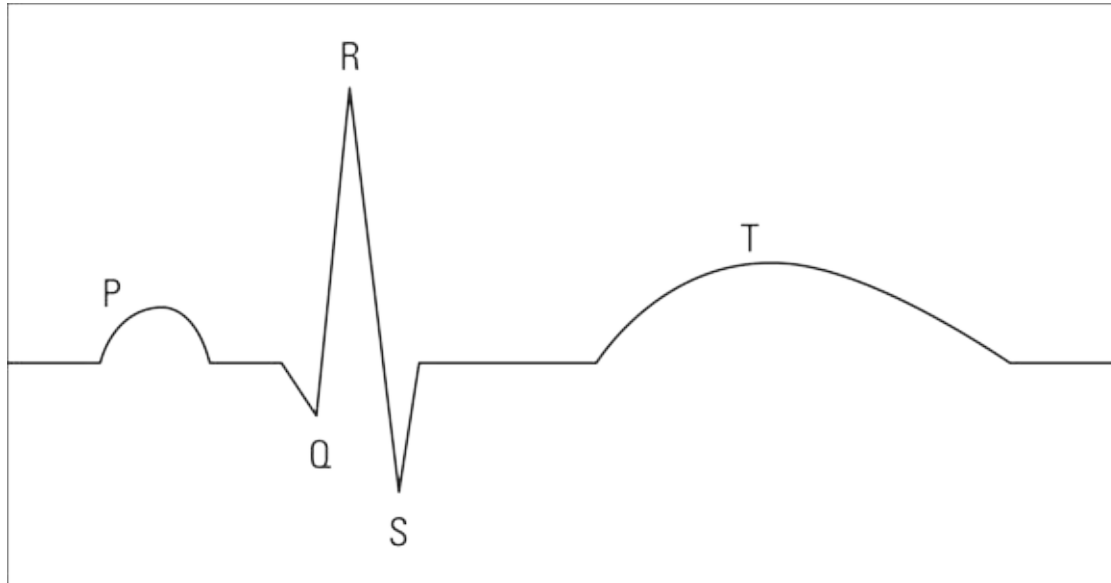


Illustration by Kathryn Born, MA

FIGURE 11-5: A typical EKG.

Riding the Network of Blood Vessels



REMEMBER Blood vessels come in three varieties, which you can see illustrated in [Figure 11-6](#):

- » **Arteries:** *Arteries* carry blood away from the heart; all arteries, except the pulmonary arteries, carry oxygenated blood. The largest artery is the aorta. Small ones are called *arterioles*, and these send off branches called *metarterioles* that join with capillaries.
- » **Veins:** *Veins* carry blood toward the heart; all veins except the pulmonary veins contain deoxygenated blood. Small ones are called *venules*, into

which the capillaries drain.

- » **Capillaries:** Microscopically small *capillaries* carry blood from arterioles to venules. These thin-walled vessels are the site of *capillary exchange*, where tissues receive resources from the blood.

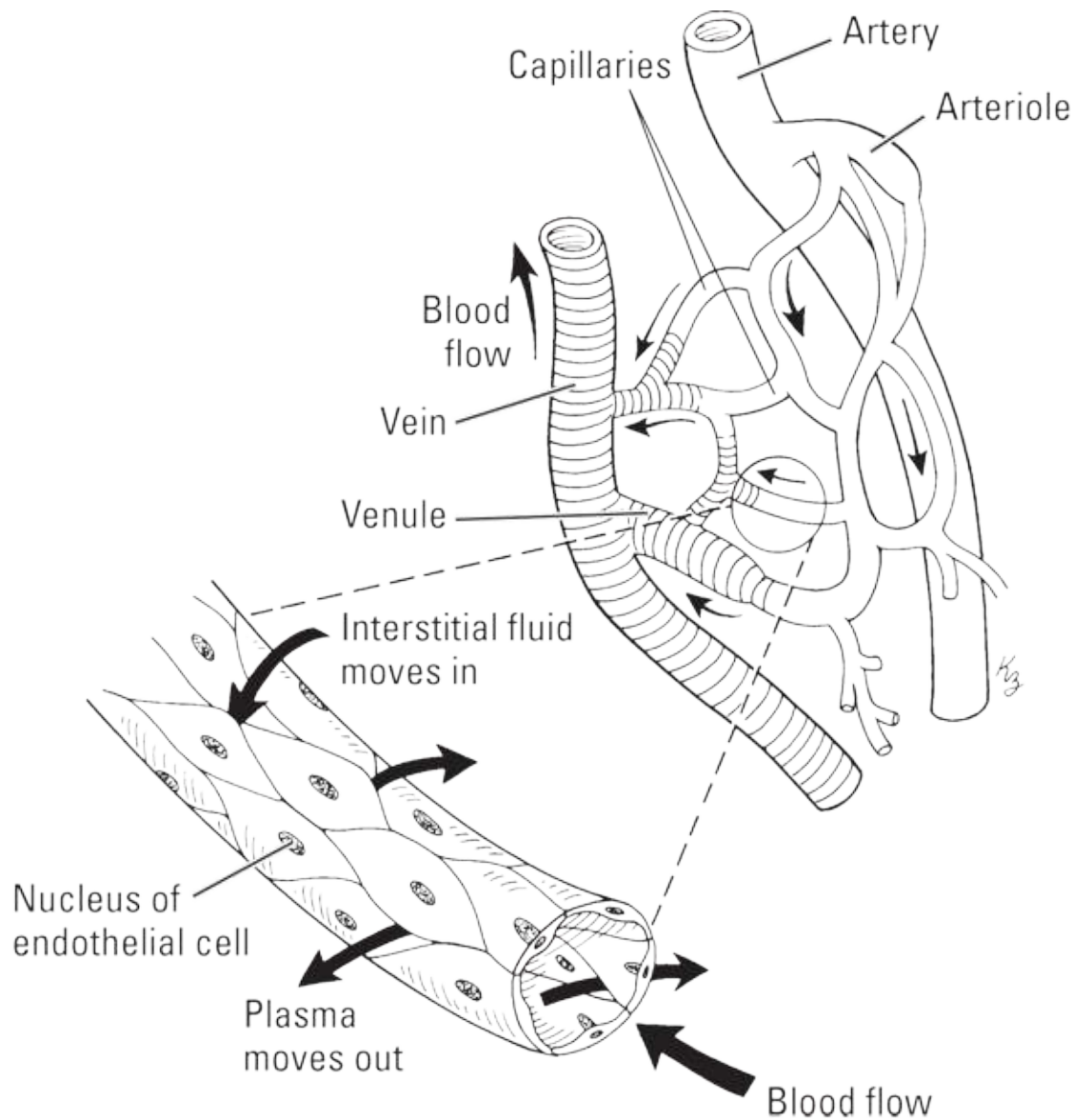


Illustration by Kathryn Born, MA

FIGURE 11-6: The capillary exchange.

The walls of arteries and veins have three layers (see [Figure 11-7](#)):

- » The outermost *tunica externa* with numerous collagen fibers to anchor the vessel to surrounding structures composed of connective tissue.

- » A central “active” layer called the *tunica media* composed of smooth muscle fibers and elastic connective tissue.
- » An inner layer called the *tunica intima* made up of simple squamous endothelium is continuous through all blood vessels. Cells of this layer secrete chemicals to prevent blood coagulation. Additionally, they regulate blood flow by influencing level of contraction in the tunica media.

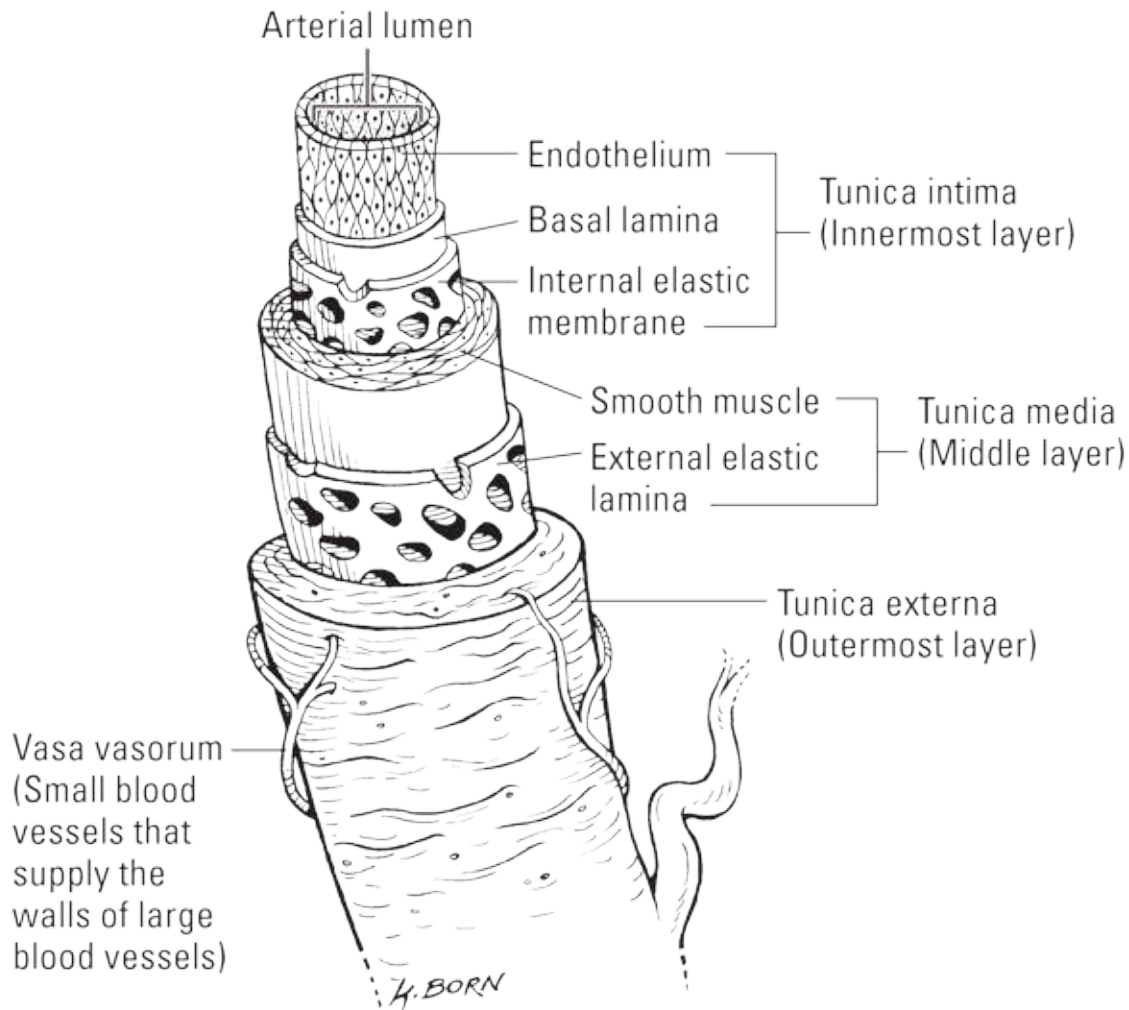


Illustration by Kathryn Born, MA

FIGURE 11-7: The anatomy of a muscular artery.

Arterial walls are very strong, thick, and very elastic to withstand the great pressure to which the arteries are subjected from the blood pushing through them.

Although larger in diameter, veins have thinner walls and are less elastic. Since the push of the blood from the heart's contraction is gone by the time it enters the veins, many of them have valves to prevent backflow. While a pressure gradient still exists (lower in the heart) it's not enough to keep the blood flowing in that direction — especially against gravity. Muscle contractions from everyday movements squeeze on the veins, pushing blood forward and then they relax, pulls blood into the (temporarily) empty space — much like how an eyedropper works.

Withstanding the pressure

You've likely had your *blood pressure* taken before — the strap wrapped around your upper arm then inflated just to the point that you consider ripping it off from the pain. The purpose of this contraption is to block blood flow to your forearm. Then, as air is let out, blood begins to flow turbulently creating the audible *sounds of Kortokoff* (that's why the stethoscope is positioned on your antecubital region, or inner elbow). When the pressure around your arm matches the *systolic blood pressure*, the maximum force that the blood puts on the walls of the arteries due to ventricular contraction, some blood gets through, hitting the walls which you can hear. Then, when the external pressure matches the *diastolic blood pressure*, the force applied to the walls between contractions, the sounds stop because the blood is now allowed to flow smoothly. So your blood pressure is recorded as the systolic pressure over the diastolic, or 120/80 mmHg on average.



REMEMBER The importance of blood pressure is not only to keep the blood circulating; it is the driving force behind capillary exchange, which is discussed in the next section.

Blood pressure that is too high (*hypertension*) damages the artery walls leading to a cascade of problems that can ultimately lead to heart failure. Unfortunately for us, the body doesn't seem to care much about this issue, as plenty of resources are being exchanged at the capillaries. There is no built-in mechanism for homeostasis of blood pressure if it is chronically high. On the other hand, low blood pressure (*hypotension*) means the body's tissues aren't receiving enough oxygen and other nutrients — a problem that must be solved post haste. A series of hormones are released via the *renin-angiotensin*

system, which leads to *vasoconstriction* (decreasing the diameter of the arteries) and water retention in the kidneys, both of which lead to an increase in blood pressure.

Capillary exchange

You may have found yourself wondering. “Just how exactly do the tissues get the resources if the blood never leaves the vessels?” Well first, you must understand that blood is a tissue composed of red blood cells, white blood cells, platelets, and plasma. Some of the plasma, which is mostly water, can exit leaving the rest of the tissue behind. This occurs easily in the capillaries because their entire wall is just one layer of simple squamous epithelium. Though the cells are held tightly together, there is still room for the plasma to escape between them — especially with the force of blood pressure. This process, called *filtration*, allows molecules in the blood to be pushed out into the *interstitial fluid* of the tissues. Cells can then bring in those necessary materials.

In addition to filtration, osmosis and diffusion play a vital role in capillary exchange. Oxygen diffuses along its concentration gradient into the interstitial fluid while carbon dioxide diffuses into the blood. This also moves other nutrients out and wastes in. Due to the outflow of water in the first half of the capillary, the water of the interstitial fluid is drawn back in, along with anything dissolved in it.

Now is your chance to practice circulating through the circulatory system:

- 48** Number the structures in the correct sequence of blood flow from the heart to the radial artery for pulse. Start at the heart with the aortic semilunar valve.

_____ Axillary artery
_____ Subclavian artery
_____ Ascending aorta
_____ Brachial artery
_____ Aortic arch

- 49** Number the structures in the correct sequence of blood flow from the great saphenous vein back to the heart.

- _____ External iliac vein
- _____ Right atrium
- _____ Common iliac vein
- _____ Femoral vein
- _____ Inferior vena cava

50 What are the sounds of Kortokoff?

- a. the pulsation of the blood on the walls of the veins during normal blood flow
- b. the sound of turbulent blood hitting the arteries when blood flow is restricted
- c. the sounds of contraction of the atria and ventricles
- d. the lub-dub sounds of the heart's valves closing

51 True or False: The renin-angiotensin system is activated in order to decrease blood pressure when it begins damaging arterial walls.

52 Which process does NOT play a major role in capillary exchange?

- a. diffusion
- b. osmosis
- c. filtration
- d. endocytosis
- e. all of the above play a role

Answers to Questions on the Cardiovascular System

The following are answers to the practice questions presented in this chapter.

- ① The system for gaseous exchange in the lungs: **b. Pulmonary circuit**
- ② The system for maintaining a constant internal environment in other

tissues: **c. Systemic circuit**

- 3 The membranous sac that surrounds the heart: **a. Pericardium**
- 4 The wall that divides the heart into two cavities: **e. Septum**
- 5 The layer providing the heart's wall with nutrients: **d. Epicardium**
- 6 What's the difference between the pulmonary and systemic circuits? **e. The pulmonary circuit travels to and from the lungs, and the systemic circuit keeps pressure and flow up for the body's tissues.** The Latin *pulmon*– means “lung.” None of the other answers even mentions the lungs.
- 7 True or False: The heart is centrally located in the chest. **False.** Actually, two-thirds of the heart lies to the left of the body's midline.
- 8 Which blood vessel contains oxygenated blood: **b. pulmonary vein.** It's the only vein in the body that does so.
- 9 Only one of the heart wall's three layers contains muscle tissue. Which one? **c. Myocardium.** *Myo*– means “muscle,” *epi*– is “upon,” *endo*– is “within,” and *peri*– is “around.”
- 10-25 Following is how [Figure 11-2](#), the heart, should be labeled.
10. **c. Brachiocephalic trunk**; 11. **j. Superior vena cava**; 12. **d. Right pulmonary veins**; 13. **p. Right atrium**; 14. **g. Right coronary artery**; 15. **n. Inferior vena cava**; 16. **k. Left common carotid artery**; 17. **f. Left subclavian artery**; 18. **h. Aortic arch**; 19. **l. Left pulmonary arteries**; 20. **o. Pulmonary trunk**; 21. **a. Left pulmonary veins**; 22. **m. Left atrium**; 23. **i. Great (left) cardiac vein**; 24. **b. Left ventricle**; 25. **e. Right ventricle**
- 26-29 Following is how [Figure 11-3](#), the heart valves, should be labeled.
26. **b. Pulmonary valve**; 27. **c. Aortic valve**; 28. **a. Tricuspid valve**; 29. **d. Mitral valve**
- 30 Why is the myocardium of the ventricles thicker than the myocardium of the atria? **c. The myocardium is the muscular tissue of the heart, and more force is needed in chambers pumping blood greater distances.** The

correct answer is the only one that correctly identifies myocardium as muscle tissue. Remember: *myo*– means “muscle.”

- 31) The superior vena cava enters the heart by way of the **d. right atrium.**
- 32) What holds the cusps of the atrioventricular valves in place? **b. The chordae tendineae.** While the papillary muscles aid in the process, it's the chordae tendineae that keep them from flapping up into the atrium.
- 33) What covers the atrioventricular opening between the right atrium and the right ventricle? **b. tricuspid valve.** Sorry if this seemed to be a trick question, but even if you have trouble remembering the heart's right openings from its left ones, you simply need to remember that the bicuspid and the mitral valve are the same thing, so “tricuspid valve” is the only correct answer here.
- 34) Returns blood to the heart from the head, thorax, and upper extremities: **c. Superior vena cava**
- 35) Valve located between the right atrium and right ventricle: **a. Tricuspid valve**
- 36) Valve located between the right ventricle and pulmonary trunk: **d. Pulmonary valve**
- 37) Returns blood to the heart from the trunk and lower extremities: **e. Inferior vena cava**
- 38) Valve located between the left atrium and left ventricle: **b. Mitral valve**
- 39) Semilunar valve located in the left ventricle: **f. Aortic valve**
- 40) Returns blood from the lungs: **g. Pulmonary vein**
- 41) Why is the sinoatrial node (SA node) called the heart's “pacemaker”? **b. It initiates an electrical impulse that causes atrial walls to contract simultaneously.** The other answers address rhythm, but the correct answer shows how the S-A node coordinates the heart's contractions.
- 42) True or False: The chordae tendineae open the AV valves. **False.** While it

looks that way since they're attached to the underside of the valve, they do not pull them open. They keep them closed; more specifically, they keep the flaps of the AV valves from being pulled up into the atrium.

- 43) Choose the correct conductive pattern. **c. S-A node → A-V node → bundle of His → bundle branches → Purkinje fibers.**
- 44) Which valve action results from the pressure in the ventricle being lower than that of the atria: **c. AV valves open**
- 45) Draw brackets on the EKG in [Figure 11-5](#) to show atrial and ventricular systole. **The bracket for atrial systole should begin at the peak of the first, smaller wave and stop at the peak of the second, larger one. The bracket for ventricular systole should begin near where the second wave reaches its peak and end where the third wave returns to baseline.**



REMEMBER The waves on an EKG do not directly show contraction. The first two show the depolarization of the fibers (spread of impulse) that triggers the contractions. The third wave shows the repolarization of the ventricles.

- 46) True or False: The atrial syncytium is comprised of the conducting fibers of the atrium that ensure the atria contract before the ventricles. **False.** While the first part of the statement is true, the purpose is to make both atria contract at the exact same time. It's the AV node that ensures the delay.
- 47) Starting with a drop of blood returning to the heart from the legs, trace its path until its eventual return to the legs. **Inferior vena cava → right atrium → tricuspid valve → right ventricle → pulmonary valve → pulmonary trunk → pulmonary arteries → lungs → pulmonary veins → left atrium → mitral valve → left ventricle → aortic valve → aorta → body**



REMEMBER To be fair, the odds of that particular drop of blood making it back to

the same place in the legs on its next trip are pretty slim. But the important point here is that it must pass through the heart twice to get sent back out to the body.

- 48) Number the structures in the correct sequence of blood flow from the heart to the radial artery for pulse. Start at the heart with the aortic semilunar valve. **1. Ascending aorta; 2. Aortic arch; 3. Subclavian artery; 4. Axillary artery; 5. Brachial artery.**
- 49) Number the structures in the correct sequence of blood flow from the great saphenous vein back to the heart. **1. Femoral vein; 2. External iliac vein; 3. Common iliac vein; 4. Inferior vena cava; 5. Right atrium.**
- 50) What are the sounds of Kortokoff : **b. the sound of turbulent blood hitting the arteries when blood flow is restricted**
- 51) True or False: The renin-angiotensin system is activated in order to decrease blood pressure when it begins damaging arterial walls: **False.** This is utilized to increase blood pressure.
- 52) Which process does NOT play a major role in capillary exchange: **d. endocytosis.** Endocytosis is the active transport mechanism cells use to bring in large molecules. Cells of the tissues would likely utilize this to bring in some of the nutrients the blood carried. It's not used to get them out of the blood; that would just put them in the cells of the capillary wall.

Chapter 12

Keeping Up Your Defenses: The Lymphatic System

IN THIS CHAPTER

- » Delving into lymphatic ducts
 - » Noodling around with nodes
 - » Exploring the lymphatic organs
-

You see it every rainy day — water, water everywhere, rushing along gutters and down storm drains into a complex underground system that most would rather not give a second thought. Well, it's time to give hidden drainage systems a second thought: Your body has one, called *the lymphatic system*. This chapter explains what you need to know about this system.

Duct, Duct, Lymph

You already know that the body is made up mostly of fluid. *Interstitial or extracellular fluid* moves in and around the body's tissues and cells constantly. The story of the lymphatic system (shown in [Figure 12-1](#)) begins deep within the body's tissues at the farthest reaches of blood capillaries, where plasma leaks out of blood capillaries at the rate of nearly 51 pints a day, carrying various substances to and away from the smallest nooks and crannies. About 90 percent of that fluid gets reabsorbed into blood capillaries taking CO₂ and other waste molecules with it. But one or two liters of extra fluid remains in the tissues as interstitial fluid. However, in order for the body to maintain a sufficient volume of water within the cardiovascular system, eventually this interstitial fluid must get back into the blood. Additionally, the fluids' contents need to be managed for the homeostasis of our internal environment. That's where the lymphatic system steps in, acting as a recycling system to drain, transport, cleanse, and return this fluid to the bloodstream.

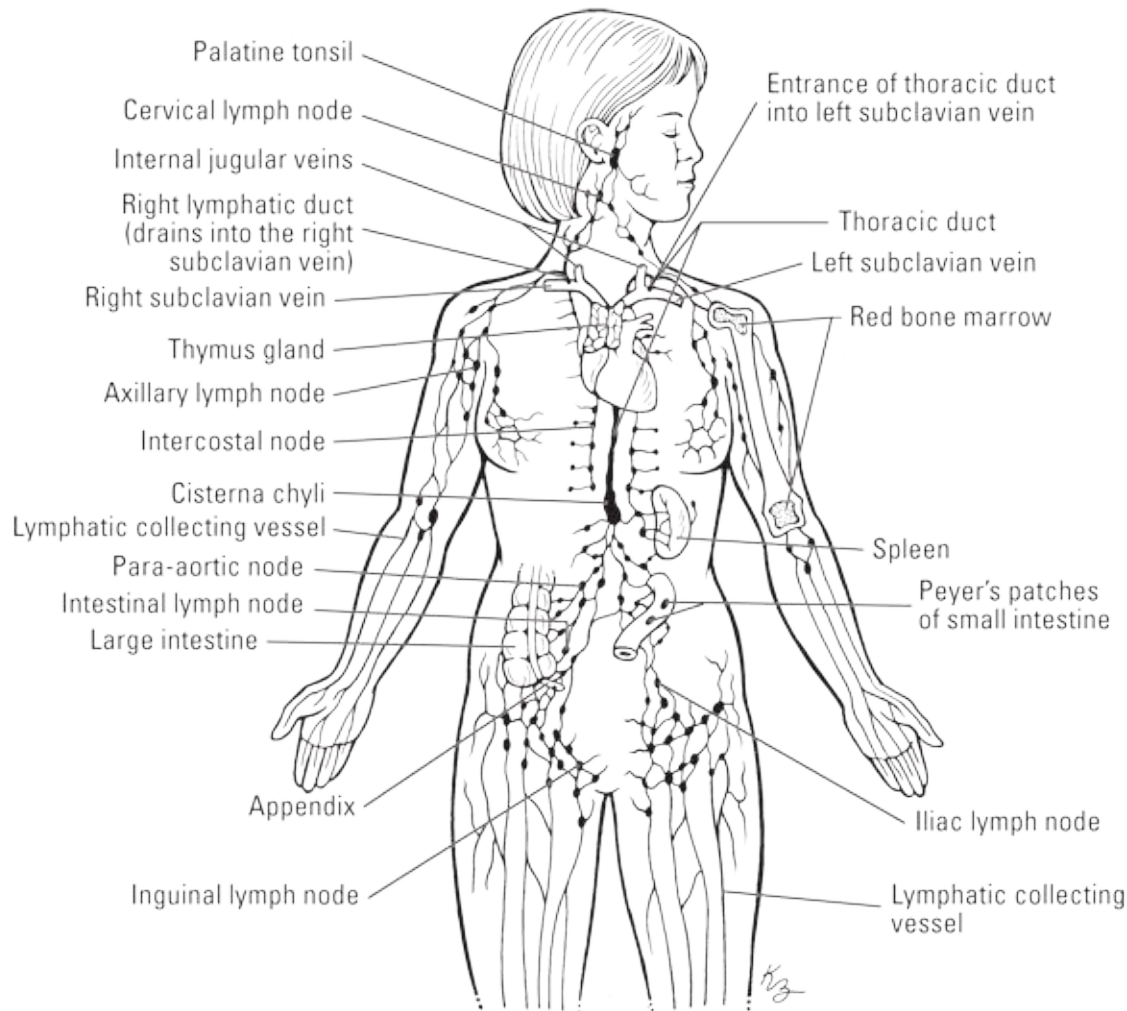


Illustration by Kathryn Born, MA

FIGURE 12-1: The lymphatic system.



REMEMBER The lymphatic system is more than a drainage network. It's a body-wide filter that traps and destroys invading microorganisms as part of the body's immune response network. It can remove pathogens from the body, help absorb and digest excess fats, and maintain a stable blood volume despite varying environmental stresses. Without it, the cardiovascular system would grind to a halt.

To collect the fluid, minute vessels called *lymph capillaries* are woven throughout the body, with a few caveats and exceptions. There are no lymph capillaries in the teeth, outermost layer of the skin, certain types of cartilage,

any other avascular tissue, or bones. And because bone marrow makes lymphocytes, which we explain in the next section, it's considered part of the lymphatic system. Plus, *lacteals* (lymphatic capillaries found in the villi of the intestines) absorb fats to mix with lymph, forming a milky fluid called *chyle*. (See [Chapter 14](#) for details on lacteals.)



WARNING It was long thought that the CNS (central nervous system) did not contain any lymphatic vessels. However, in 2017, researchers were finally able to use advances in imaging technology to confirm that there are, in fact, lymphatic vessels in and around the brain.

Unlike blood capillaries, lymph capillaries dead-end (terminate) within tissue. Made up of a single layer of loosely overlapping endothelial cells anchored by fine filaments, lymph capillaries behave as if their walls are made of cellular, one-way valves. Endothelium provides a smooth, slick, friction-free surface in the capillaries. When the pressure outside a capillary is greater than it is inside, the filaments anchoring the cells allow them to open, permitting interstitial fluid to seep in. Increasing pressure inside the capillary walls eventually forces the entry gates to close. Once in the capillaries, the trapped fluid is known as *lymph*, and it moves into larger, vein-like *lymphatic vessels*. The lymph moves slowly and without any kind of central pump through a combination of peristalsis, the action of semilunar valves, and the squeezing influence of surrounding skeletal muscles, much like that which occurs in veins.



TIP Plasma, interstitial fluid, and lymph are all basically the same thing — body fluid made mostly of water carrying various molecules. The difference is location. If the fluid is in the blood, then we call it plasma. Out in the tissues? Now it's interstitial fluid. Located in a lymphatic vessel? Presto! Now it's lymph.

Lymphatic capillaries are found in conjunction with blood capillaries and the vessels form networks around blood vessels. The larger lymphatic vessels contain *lymph nodes*, which are bean-shaped filters (more on these in the next

section). The vessels then merge into *lymphatic trunks*, which merge into either the *thoracic duct* or the *right lymphatic duct*. The trunks are named for the region they serve (lumbar, intestinal, intercostal, subclavian, jugular, bronchomediastinal). Both ducts drain the lymph into the respective subclavian veins (thoracic goes into the left subclavian), where it becomes plasma again.

To see how much of this information is seeping in, answer the following questions:

- 1 The lymphatic system plays an important role in regulating
 - a. intracellular energy function.
 - b. interstitial fluid protein.
 - c. metabolizing unused fats from the small intestine.
 - d. intercellular transportation of oxygen.
- 2 Under what circumstances do the walls of lymphatic capillaries open?
 - a. When the filaments holding the cells together begin to dehydrate
 - b. When the pressure outside the capillaries is greater than inside
 - c. When the capillaries sense extra salts in the bloodstream
 - d. When reticular fibers begin to force their way between the cells
- 3 How does lymph move around inside its vessels?
 - a. Rapidly, through the action of tiny pumps called lymph nodes
 - b. Slowly, and only as additional lymph enters the system
 - c. Slowly, through peristalsis, the action of semilunar valves, and the squeezing of surrounding skeletal muscles
 - d. Steadily and rhythmically, mirroring the heart rate
- 4 What is the largest lymphatic vessel in the body?
 - a. Right lymphatic duct
 - b. Spleen

- c. Thoracic duct
- d. Lymph node
- e. Bone marrow

5 The lymphatic system serves several crucial functions, but what does it NOT normally do?

- a. Return interstitial fluid to the blood
- b. Destroy bacteria
- c. Remove old erythrocytes
- d. Filter the lymph
- e. Produce erythrocytes

Poking at the Nodes

Lymph nodes (see [Figure 12-2](#)) are the site of filtration in the lymphatic system. Also sometimes incorrectly referred to as *lymph glands* — they don't secrete anything, so technically they're not glands — these bean-shaped sacs are surrounded by connective tissue (and therefore are tough to spot). Lymph nodes contain macrophages, which destroy bacteria, cancer cells, and other matter in the lymph fluid. *Lymphocytes* (a type of white blood cell), which produce an immune response to microorganisms, also are found in lymph nodes. The indented part of each node, called the *hilum*, is where the efferent vessels exit and where the blood vessels (that supply the node tissue) enter and exit. Afferent vessels bring the lymph in on the convex side. The *stroma* (body) of each node is surrounded by a fibrous capsule that dips into the node to form *trabeculae*, or *septa* (thin dividing walls) that divide the node into compartments. *Reticular* (netlike) fibers are attached to the trabeculae and form a framework for the lymphoid tissue and clusters of macrophages and B lymphocytes called *lymphatic nodules*.

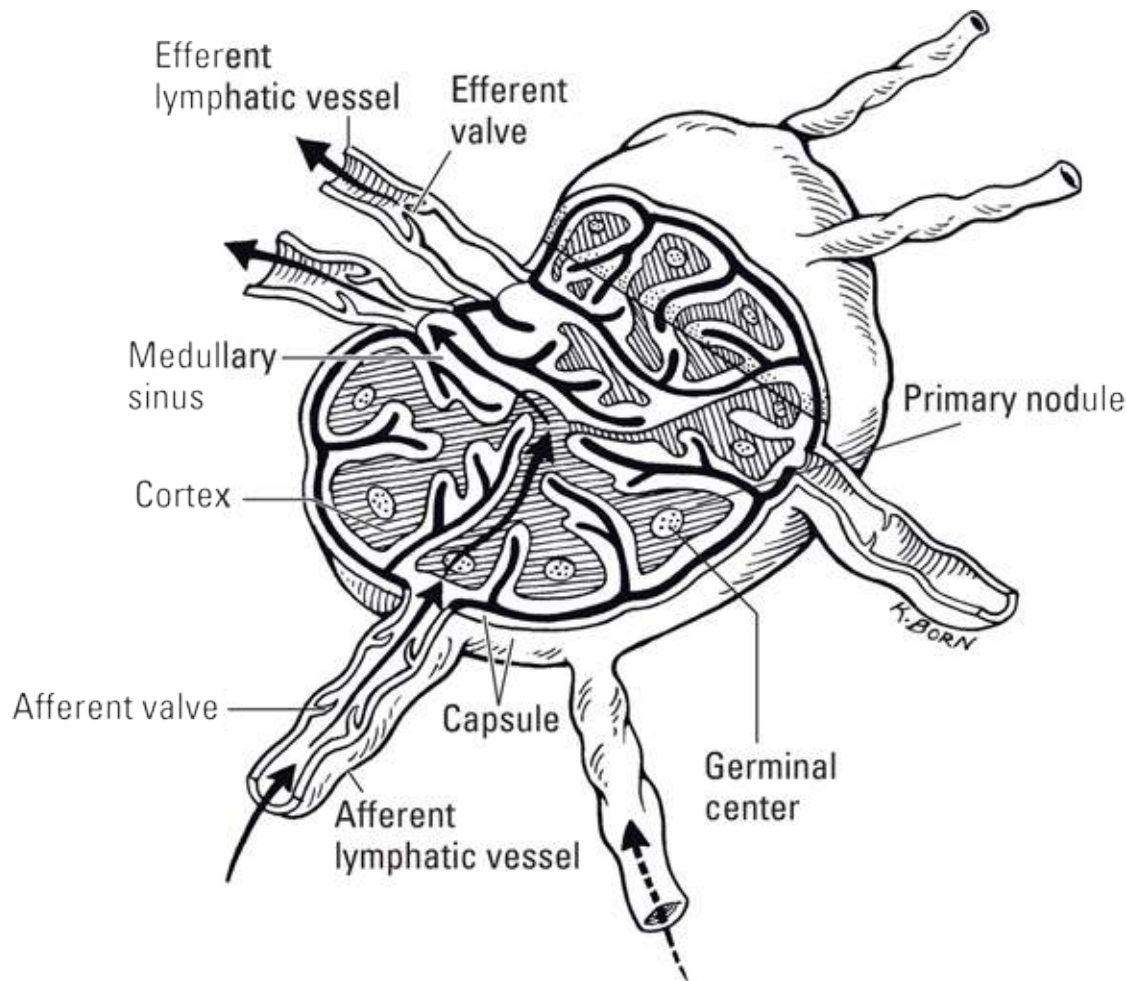


Illustration by Kathryn Born, MA

FIGURE 12-2: A lymph node.



TIP If you have trouble remembering your afferent from your efferent, think of the “a” as standing for “access” and the “e” as standing for “exit.”

Although some lymph nodes are isolated from others, most nodes occur in groups, or clusters, particularly in the inguinal (groin), axillary (armpit), and mammary gland areas. (You can see some lymph nodes in [Figure 12-1](#).) The following are the primary lymph node regions:

- » **Cervical:** Found in the neck, filter lymph from the head
- » **Axillary:** Found in armpits, filter lymph from arms and mammary region

- » **Supratrochlear:** Found above inner elbow, filter lymph from hands
- » **Inguinal:** Found in inguinal region, filter lymph from lower limbs and external genitalia
- » **Pelvic:** Found in pelvic cavity, filter lymph from pelvic organs
- » **Abdominal:** Found in abdominal cavity, filter lymph from abdominal organs
- » **Thoracic:** Found in mediastinum, filter lymph from heart and lungs

Each node acts like a filter bag filled with a network of thin, perforated sheets of tissue — a bit like cheesecloth — through which lymph must pass before moving on. White blood cells line the sheets of tissue, including several types that play critical roles in the body's immune defenses. This filtering action explains why, when infection first starts, lymph nodes often swell with the cellular activity of the immune system launching into battle with the invading microorganisms.

Think you have a node-tion (sorry!) about what's happening here? Test your knowledge:



EXAMPLE Q. An area of the body where no lymph nodes are found is the

- a. integument.
- b. liver.
- c. stomach.
- d. bones.

A. The correct answer is bones. Though bone marrow is considered part of the lymphatic system, the bones do not contain any lymph nodes.

6-7 Fill in the blanks to complete the following sentence.

Lymph moves from a/an **6.** _____ vessel through a lymph node and back out into the blood through a/an **7.** _____ vessel.

8 What do lymph nodes use to destroy bacteria, cancer cells, and other foreign material?

- a. Stroma
- b. Trabeculae
- c. Peyer's patches
- d. Macrophages

9 Why do lymph nodes sometimes swell when an infection is present?

- a. To make room for additional lymph fluid
- b. Because of increased viscera that results from gastric activity
- c. To accommodate cellular activity of the immune-system-attacking microorganisms
- d. Because white blood cells are larger than red blood cells

10 Lymph nodes have two primary functions. What are they?

- a. To conserve white blood cells and to produce lymphocytes
- b. To produce bilirubin and to cleanse lymph fluid
- c. To produce lymphocytes and antibodies and to filter lymph fluid
- d. To conserve iron and to remove erythrocytes
- e. To produce erythrocytes and to conserve iron

11 The two lymphatic ducts empty into the

- a. lymph nodes
- b. spleen
- c. thymus
- d. right atrium
- e. subclavian veins

12 The germinal center of lymph nodules produces what type of cells?

- a. Monocytes
- b. Lymphocytes
- c. Phagocytes
- d. Trabeculae
- e. Erythrocytes

13 The connective tissue fiber that forms the framework of the lymphoid tissue is

- a. cartilaginous.
- b. collagenous.
- c. reticular.
- d. elastic.

14 Which of the following regions of lymph nodes filter lymph from the lower limbs?

- a. axillary
- b. pelvic
- c. cervical
- d. inguinal
- e. abdominal

Having a Spleen-did Time with the Lymphatic Organs

Although the lymph nodes are the most numerous lymphatic organs, several other vital organs exist in the lymphatic system, including the spleen, thymus gland, and tonsils.

Reuse and recycle: The spleen

The spleen, the largest lymphatic organ in the body, is a 5-inch, roughly egg-shaped structure to the left of and slightly behind the stomach. Like lymph

nodes, it has a hilum through which the splenic artery, splenic vein, and efferent vessels pass. (The efferent vessels transport lymph away from the spleen to lymph nodes; as we note in the preceding section, remember “e” for “exit.”) Also like lymph nodes, the spleen is surrounded by a fibrous capsule that folds inward to section it off. Arterioles leading into each section are surrounded by masses of developing lymphocytes that give those areas of so-called *white pulp* their appearance. On the outer edges of each compartment, tissue called *red pulp* consists of blood-filled cavities. Unlike lymph nodes, the spleen doesn’t have any afferent (access) lymph vessels, which means that it doesn’t filter lymph, only blood.

Blood flows slowly through the spleen to allow it to remove microorganisms, exhausted erythrocytes (red blood cells), and any foreign material that may be in the stream. Among its various functions, the spleen can be a blood reservoir. When blood circulation drops while the body is at rest, the spleen’s vessels can dilate to store any excess volume. Later, during exercise or if oxygen concentrations in the blood begin to drop, the spleen’s blood vessels constrict and push any stored blood back into circulation.



REMEMBER The spleen’s primary role is as a biological recycling unit, capturing and breaking down defective and aged red blood cells to reuse their components. Iron stored by the spleen’s macrophages goes to the bone marrow, where it’s turned into hemoglobin in new blood cells. By the same token, bilirubin for the liver is generated during breakdown of hemoglobin. The spleen produces red blood cells during embryonic development but shuts down that process after birth; in cases of severe anemia, the spleen sometimes starts up production of red blood cells again.

Fortunately, the spleen isn’t considered a vital organ; if it is damaged or has to be surgically removed, the liver and bone marrow can pick up where the spleen leaves off.

T cell central: The thymus gland

Tucked just behind the breastbone and between the lungs in the upper chest (the *superior mediastinum*, if you want to be technical), the thymus gland was

a medical mystery until recent decades. Its two oblong lobes are largest at puberty when they weigh around 40 grams (somewhat less than an adult mouse). Through a process called *involution*, however, the gland atrophies and shrinks to roughly 6 grams by the time an adult is 65. (You can remember that term as the inverse of evolution.)

The thymus gland serves its most critical role — as a nursery for immature T lymphocytes, or T cells — during fetal development and the first few years of a human's life. Prior to birth, fetal bone marrow produces *lymphoblasts* (early stage lymphocytes) that migrate to the thymus. Shortly after birth and continuing until adolescence, the thymus secretes several hormones, collectively called *thymosin*, that prompt the early cells to mature into full-grown T cells that are *immunocompetent*, ready to go forth and conquer invading microorganisms. (These hormones are the reason the thymus is considered part of the endocrine system, too; check out [Chapter 10](#) for details on this system.)

As with other lymphatic structures, the thymus is surrounded by a fibrous capsule that dips inside to create chambers called *lobules*. Within each lobule is a cortex made of T cells held in place by reticular fibers and a central medulla of unusually onion-like layered epithelial cells called *thymic corpuscles*, or *Hassall's corpuscles*, as well as scattered lymphocytes.

Open wide and move along: The tonsils and Peyer's patches

Like the thymus gland, the tonsils, which are misunderstood masses of lymphoid tissue, are largest around puberty and tend to atrophy as an adult ages. Unlike the thymus, however, the tonsils don't secrete hormones but do produce lymphocytes and antibodies to protect against microorganisms that are inhaled or eaten. Although only two are visible on either side of the pharynx, there are actually five tonsils:

- » The two you can identify, which are called *palatine tonsils*
- » The *adenoid* or *pharyngeal tonsil* in the posterior wall of the nasopharynx
- » The *lingual tonsils*, which are round masses of lymphatic tissue arranged in two approximately symmetrical collections that cover the posterior one-third of the tongue

Invaginations (ridges) in the tonsils form pockets called *crypts*, which trap bacteria and other foreign matter.

Peyer's patches are masses of lymph nodules just below the surface of the ileum, the last section of the small intestine. When harmful microorganisms get into the intestine, Peyer's patches can mobilize an army of B cells and macrophages to fight off infection.

You've absorbed a lot in this section. See how much of it is getting caught in your filters:

15 What role does the spleen have besides that of a biological recycling unit?

- a. It undergoes involution as a person ages to make way for enlarged lymph nodes.
- b. It divides the pharynx from the tonsils to ensure that lymph fluid moves in the right direction.
- c. It produces neutrophils to maintain immunocompetent T cells.
- d. It has vessels that dilate to store excess blood volume when the body is at rest.

16 Why does the thymus stop growing during adolescence and atrophy with age?

- a. It serves its most critical role during fetal development and the first few years of life.
- b. The increasing number of lymph nodes over time takes over its role in the body.
- c. The reticular fibers surrounding it contract and crush it over time.
- d. It slowly runs out of Hassall's corpuscles.

17 Lymphoid tissue(s) located in the pharynx that protect(s) against inhaled or ingested pathogens and foreign substances is/are called the

- a. thymus.
- b. tonsils.
- c. Peyer's patches.

d. spleen.

18 Lymphatic nodules found in the ileum of the small intestines are

- a. Peyer's patches.
- b. lymph nodes.
- c. thymus.
- d. macrophages.

19 What is the significance of the spleen having no afferent vessels?

- a. Foreign materials can't get inside and cause problems.
- b. It means that the spleen filters only blood, not lymph.
- c. It's why humans can survive removal of the spleen.
- d. The spleen evolved that way to prevent phagocytosis.

20 The lymphatic organ found in the superior mediastinum is the

- a. tonsil.
- b. spleen.
- c. thymus.
- d. reticular formation.

21 Why is the spleen's white pulp that color?

- a. Blood has drained from the red pulp back into the bloodstream.
- b. Cartilage is beginning to form at the borders.
- c. Lymphocytes clump there and give it its color.
- d. The elastic connective tissue has stretched and relaxed repeatedly over the course of a person's lifetime.

22 The lymphatic organ(s) responsible for removal of aged and defective red blood cells from the bloodstream is/are the

- a. tonsils.
- b. spleen.
- c. Peyer's patches.
- d. lymph nodes.

23 Why does the thymus secrete hormones?

- a. To make T lymphocytes immunocompetent
- b. To help the tonsils make more T lymphocytes
- c. To support the Peyer's patches as they produce B lymphocytes
- d. To signal the spleen that it's time to release more red blood cells

Investigating Immunity

The lymphatic system is the battleground for our immune processes. But nearly every tissue in our bodies have some role to play (which is why you no longer see the immune system as one of the body's listed systems). We still have much to learn about all the mechanisms our bodies use to defend us from invaders, but this section provides a brief overview of what we do know.

First, it's important to understand that our body's defenses are divided into two categories: innate and adaptive.

Innate defenses

Innate, or non-specific, defenses are the tools our bodies use to attack foreign invaders regardless of their ilk. Germs can be bacteria, viruses, fungi, or other microorganisms and other foreign particles (pollen, toxins) can be problematic as well. Our innate defenses target all of these.

First and foremost is our skin — the body's largest organ and our first line of defense. Along with our other mechanical barriers, such as mucus and tears, most of the potential invaders are never even allowed entry. Should one make it into the body we have other innate strategies for our second line of defense:

» Chemical barriers

- Enzymes (in saliva, gastric juice) break down cell walls.

- *Interferons* block replication (especially of virus and tumor cells).
 - *Defensins* poke large holes in cell membranes.
 - *Collectins* group together pathogens for easier phagocytosis.
- » **Inflammation:** Dilates blood vessels, sending more resources to the area where the pathogen was identified
- » **Fever:** Weakens microorganisms and stimulates phagocytosis
- » **Natural killer cells (NKs):** Secrete *perforins* to poke tiny holes in, or perforate, cell membranes
- » **Phagocytosis:** Consumption of foreign invaders by specialized white blood cells

Unfortunately, the occasional pathogen makes it past these defenses so our bodies mount a targeted attack. Furthermore, if we relied solely on our innate defenses, there would be massive amounts of collateral damage to our own cells (which is responsible for many of our symptoms of illness in the first place).

Adaptive defenses

Our *adaptive*, or *specific*, *defense* mounts a two-tiered attack targeting the specific pathogen. This minimizes collateral damage but takes time to get started. This process is dependent on molecules that stick off the surface of cells called *antigens*. All cells have them, unique to their variety, and that's how our immune cells distinguish self versus non-self. A type of white blood cell called a *macrophage* destroys a pathogen by phagocytosis; however, it leaves the antigens intact and displays them on itself. This way, it's one of our own cells that looks foreign searching for the matching lymphocytes to initiate our adaptive response.

There are two varieties of lymphocyte that carry out this response: *T cells* which mature in the thymus and *B cells* which mature in the bone marrow (see the connection?). The action of T cells is called *cell-mediated immunity* and of B cells it is called *humoral immunity*.

Cell-mediated

Once a macrophage finds a T cell with receptors that match its displayed antigens, they bind together. The lymphocyte, called a *helper T cell*, releases

a chemical called *interleukin-2*, which activates another matching T cell. This stimulates the now *cytotoxic T cell* to begin proliferating (making copies of itself). These cytotoxic Ts (sometimes called killer Ts) will bind with antigens on the invader and release perforins, killing the pathogen. So only cells with this particular antigen will be targeted. When the battle has waned, *suppressor T cells* signal the adaptive immune process to stop. Some T cells will remain as *memory T cells* once the pathogen has been defeated. This way, if it invades again, it won't take long for the macrophage to find a match and the pathogen will be destroyed before you even show any symptoms — thus providing you *immunity*.

Humoral

B cells, with matching receptors, will bind to the pathogen or the antigen-presenting macrophage. When the helper T cell is activated it also releases *cytokines* which, in turn, activate the B cell. It begins to proliferate into *plasma B cells* and *memory B cells*. The memory Bs hang around with the memory T cells in the lymph nodes for protection later. The plasma Bs begin manufacturing *antibodies*, which are proteins that will bind to the antigens on the pathogens. When bound with antibodies, the pathogen is now *neutralized*. Since they have two binding sites, antibodies can also cause *agglutination*, clumping together the invaders for more efficient phagocytosis. They also can activate the *complement cascade*, a series of chemical reactions that can directly destroy the pathogen.

The faster we can locate the matching B and T cells, the less damage the pathogen can cause. Lymphocytes are generated with random receptor shapes and researchers argue that we all have one cell in us somewhere to match any pathogen we could possibly encounter — the issue is, can we find it before the pathogen does irreversible damage.

Hopefully these questions won't seem too foreign:

- 24** Why are macrophages considered the bridge between our innate and adaptive immune responses?

25-30 Match the immune strategy to its description.

- a. block entry into body
- b. disrupt replication of pathogen
- c. gather pathogens together for destruction
- d. tear large holes in cell membrane
- e. released by NKs to poke tiny holes in cell membranes
- f. white blood cell engulfs the entire pathogen

- 25. _____ collectins
- 26. _____ defensins
- 27. _____ interferons
- 28. _____ mucus
- 29. _____ perforins
- 30. _____ phagocytosis

31 One of the mechanisms used by antibodies is to clump together pathogens for more efficient phagocytosis. This is called:

- a. complement cascade.
- b. neutralization.
- c. precipitation.
- d. agglutination.
- e. collectins.

32-36 Fill in the blanks to complete the following paragraph.

Pathogens, like our own cells, are covered in **32.**_____ which enable our white blood cells to distinguish between self and non-self. When activated, **33.**_____ that have the matching receptors release perforins to destroy the specific pathogen. Meanwhile, B cells synthesize **34.**_____ that will target the pathogen as well. Both

types of lymphocyte keep 35. _____ around for a faster response on the next exposure, granting you 36. _____ from that particular pathogen.

Answers to Questions on the Lymphatic System

The following are answers to the practice questions presented in this chapter.

- ① The lymphatic system plays an important role in regulating **b. interstitial fluid protein**. By keeping the interstitial fluid volume between tissue cells in balance, the lymphatic system also keeps the body in homeostasis.
- ② Under what circumstances do the walls of lymphatic capillaries open? **b. When the pressure outside the capillaries is greater than inside**. The difference in pressure causes the filaments anchoring the thin layer of cells to open. Increasing pressure inside the capillary walls eventually forces them closed again.
- ③ How does lymph move around inside its vessels? **c. Slowly, through peristalsis, the action of semilunar valves, and the squeezing of surrounding skeletal muscles**. You already know the first answer is wrong because lymph nodes aren't pumps, and because you know the lymphatic system doesn't have pumps, you know the last answer is wrong because there is nothing steady or rhythmic about it. You're left with one of the "slowly" answers. The second answer can't be right because the system wouldn't have room to take in more lymph if what was already there wasn't moving through some kind of action. Voilà! You have the right answer.
- ④ What is the largest lymphatic vessel in the body? **c. Thoracic duct**. Yes, this duct is the largest lymphatic vessel. "Spleen" isn't the correct answer because that's the largest lymphatic *organ*.
- ⑤ The lymphatic system serves several crucial functions, but what does it *not* normally do? **e. Produce erythrocytes**. Those are red blood cells, which develop in the bone marrow.

- 6-7 Lymph moves from a/an **6. afferent lymphatic** vessel through a lymph node and back out into the blood through a/an **7. efferent lymphatic** vessel. Not sure which vessel is which? A comes first in the alphabet, and you can think of it as the lymph's Access point. E comes next, and it provides an Exit point.
- 8 What do lymph nodes use to destroy bacteria, cancer cells, and other foreign material? **d. Macrophages.** You can arrive at the correct answer through a process of elimination. Neither stroma nor trabeculae can be right because you already know that they're both part of connective tissue. Peyer's patches are limited to the small intestine. So macrophages must be the front-line warriors here.
- 9 Why do lymph nodes sometimes swell when an infection is present? **c. To accommodate cellular activity of the immune-system-attacking microorganisms.** You can eliminate a couple of answers right off the bat here. The first answer can't be right; why would there be a need for additional lymph fluid? You also know the word "gastric" in the second answer relates to digestion, not the lymphatic system.
- 10 Lymph nodes have two primary functions. What are they? **c. To produce lymphocytes and antibodies and to filter lymph fluid.** Be careful that both parts of the answer are correct.
- 11 The two lymphatic ducts empty into the **e. subclavian veins.**
- 12 The germinal center of lymph nodules produces what type of cells? **b. Lymphocytes.** Your answer hint is in the question: *Lymph* nodules produce *lymphocytes*.
- 13 The connective tissue fiber that forms the framework of the lymphoid tissue is **c. reticular.** It provides both a tissue framework and a type of netting to hold clusters of lymphocytes.
- 14 Which of the following regions of lymph nodes filter lymph from the lower limbs? **d. inguinal.**
- 15 What role does the spleen have besides that of a biological recycling unit? **d. It has vessels that dilate to store excess blood volume when the body**

is at rest. Enlarged lymph nodes don't travel, so you know the first answer is wrong. Plus, you know the spleen is in the abdomen, not the throat, so toss out that second answer. Wavering between the third and fourth answers? Just remember all those TV shows where internal bleeding is chalked up to a damaged spleen, and you've got it!

- 16) Why does the thymus stop growing during adolescence and atrophy with age? **a. It serves its most critical role during fetal development and the first few years of life.**



TIP Here's a memory tool that only word-play students will love: "The thymus runs out of thyme."

- 17) Lymphoid tissue(s) located in the pharynx that protect(s) against inhaled or ingested pathogens and foreign substances is/are called the **b. tonsils**. When you remember that the pharynx is the throat, this question becomes more obvious.
- 18) Lymphatic nodules found in the ileum of the small intestines are **a. Peyer's patches**. It's almost like they're "patched" onto the ileum.
- 19) What is the significance of the spleen having no afferent vessels? **b. It means that the spleen filters only blood, not lymph.** No afferent vessels means no access for lymph to get in there.
- 20) The lymphatic organ found in the superior mediastinum is the **c. thymus**.



TIP Break this question into parts and it becomes easier to locate which gland is being referenced: *Superior* means "upper," *media-* means "middle" (or "midline"), and *-stinum* refers to the sternum, or breastbone.

- 21) Why is the spleen's white pulp that color? **c. Lymphocytes clump there and give it its color.** Red pulp is always red, so say goodbye to that first answer. The spleen doesn't need cartilage, so toss out the second answer.

And the final answer can't be right because what would be the evolutionary point in scarring up the spleen as a person ages?

- 22) The lymphatic organ(s) responsible for removal of aged and defective red blood cells from the bloodstream is/are the **b. spleen**. It recycles critical components from the spent red blood cells and sends them to the bone marrow to be turned into fresh cells.
- 23) Why does the thymus secrete hormones? **a. To make T lymphocytes immunocompetent**. Plus, it's where these cells get the "T" in their name.
- 24) Why are macrophages considered the bridge between our innate and adaptive immune responses? **Macrophages are, themselves, an innate defense. They consume anything with foreign antigens so it's non-specific. However, it then displays those antigens trying to find a helper T with matching receptors to activate in order get the adaptive response going.**
- 25) collectins **c. gather pathogens together for destruction**
- 26) defensins **d. tear large holes in cell membranes**
- 27) interferons **b. disrupt replication of pathogen**
- 28) mucus **a. block entry into body**
- 29) perforins **e. released by NKs to poke tiny holes in cell membranes**
- 30) phagocytosis **f. white blood cell engulfs the entire pathogen**
- 31) One of the mechanisms used by antibodies is to clump together pathogens for more efficient phagocytosis. This is called **d. agglutination**. Watch out for collectins; they serve the same purpose, but this is an innate chemical defense produced by white blood cells.
- 32-36) Pathogens, like our own cells, are covered in **32. antigens** which enable our white blood cells to distinguish between self and non-self. When activated, **33. Cytotoxic T cells** that have the matching receptors release perforins to destroy the specific pathogen. Meanwhile, B cells synthesize **34. antibodies** that will target the pathogen as well. Both types of

lymphocyte keep **35. memory cells** around for a faster response on the next exposure, granting you **36. immunity** from that particular pathogen.

Chapter 13

Oxygenating the Machine: The Respiratory System

IN THIS CHAPTER

- » Breathing in the basics
 - » Identifying the organs and muscles of the respiratory tract
 - » Getting to know gas exchange
-

People need lots of things to survive, but the most urgent need from moment to moment is oxygen. Without a continual supply of this vital element, we don't last long. But if we have reserves of the other things we need — carbohydrates, fats, and proteins — why don't we have some kind of storehouse of oxygen, too? Simple. It's readily available in the air around us, so we've never needed to evolve a means for storing it. Nonetheless, our stored food supplies would be useless without oxygen; our bodies can't metabolize the energy they need from these substances without a constant stream of oxygen to keep things percolating along.

All that metabolizing creates another equally important need, however. We must have a means for getting rid of our bodies' key gaseous waste — carbon dioxide, or CO₂. If it builds up in our systems, we die. It must be removed from our bodies almost as fast as it's formed. Conveniently, breathing in fulfills our need for oxygen and breathing out fulfills our need to expel carbon dioxide.

In this chapter, you get a quick review of Mother Nature's dual-purpose system and plenty of opportunities to test your knowledge about the lungs and other parts of the respiratory system.

Breathing In the Basics



REMEMBER *Respiration*, or the exchange of gases between an organism and its environment, occurs in three distinct processes:

- » **Breathing:** The technical term is *pulmonary ventilation*, or the movement of air into and out of the lungs. Breathing is comprised of two distinct actions: *inspiration* and *expiration*.
- » **Exchanging gases:** This takes place between the lungs, the blood, and the body's cells in two ways:
 - **Pulmonary, or external, respiration:** The exchange in the lungs when blood gains oxygen and loses carbon dioxide
 - **Systemic, or internal, respiration:** The exchange that takes place in and out of capillaries when the blood releases some of its oxygen and collects carbon dioxide from the tissues
- » **Cellular respiration:** Oxygen is used in the catabolism of substances like glucose for the production of energy (see [Chapter 2](#)), creating CO₂ as a byproduct.

A single respiratory cycle consists of one inhalation followed by an expiration. The regular, restful breathing rate is controlled by the pons, while the medulla oblongata will signal any necessary changes to that pattern. To complete a normal inhalation, the *diaphragm* (the broad skeletal muscle that forms the bottom of the thoracic cavity) is triggered to contract. This pushes down on the contents of the abdominal cavity, thus increasing the volume of the lungs. An increase in volume causes a decrease in pressure (known as *Boyle's Law*). So, as a result of the diaphragm contracting, the pressure of the air already inside the lungs drops below that of atmospheric pressure (the pressure of the air outside our bodies). Because gasses will naturally diffuse to areas of lower pressure, air flows into the lungs. Thus we do not need to “suck in” air with each normal breath. For a deeper inhalation, we contract the intercostal muscles (between the ribs), which pull the ribcage out, further increasing the volume and dropping the pressure inside the lungs even lower so more air can come in. Once inside the lungs, gasses can be exchanged between the air and the blood.

Exhalation is a passive process; that is, we don't tell the lungs to breathe out. We simply stop telling the diaphragm (and the intercostal muscles if they're engaged) to contract. When they relax, volume decreases and pressure increases. Further, the lungs contain a great deal of elastic tissue. As the muscles relax, the elastic tissue snaps back. This *elastic recoil* briefly drops the pressure inside the lungs to below atmospheric pressure and air flows out. Because we can consciously amplify our ventilation, there are differing, measurable lung volumes. These are summarized below:

- » **Tidal volume:** The amount of air moved in or out during normal ventilation
- » **Inspiratory reserve:** The additional amount of air that can be inhaled with a forceful inspiration
- » **Expiratory reserve:** The additional amount of air that can be exhaled with a forceful expiration
- » **Residual volume:** The amount of air that must remain in the lungs at all times (the lungs never deflate with exhalation, even if it's forceful)
- » **Vital capacity:** The maximum amount of air that can be moved into (or out of) the lungs
- » **Total lung capacity:** The total amount of air the lungs can hold; equal to the vital capacity plus the residual volume

Air enters the lungs through the *trachea* which branches to the right and left, forming the *primary bronchi*. These then branch several times forming the *bronchial tree* which distributes the air throughout the lungs:

Primary bronchi → Secondary bronchi → Tertiary bronchi →
Bronchioles → Alveolar ducts → Alveoli

The alveoli are the air sacs and are the site of gas exchange. The bulk of the respiratory tract is wrapped in rings of hyaline cartilage (to maintain their shape) and is lined with ciliated pseudostratified tissue. The lining contains numerous *goblet cells* to produce mucus to trap unwanted particles in the air (like dust). The cilia then sweep the mucus up and out to be expelled — sometimes with the aid of a cough. The further down in the bronchial tree you

proceed, the fewer the goblet cells you see. (We definitely don't want mucus in our air sacs.) The bronchioles lack the supportive cartilage rings and are lined with simple cuboidal, while the alveoli are comprised of simple squamous tissue.

The lungs themselves are divided into lobes: The right has three while the left has two (to leave room for the heart). Between the right and left lungs is the *mediastinum*, the central compartment of the thoracic cavity that houses the heart, major blood and lymphatic vessels, nerves, trachea, and esophagus. Each lung is covered in the *visceral pleura*. Since the lungs occupy most of the remaining space inside the thoracic cavity, there is little to no space between the visceral pleura and the *parietal pleura*, which lines the thoracic cavity (refer to [Chapter 1](#) to brush up on these layers and compartments). The attraction between the two pleural membranes facilitates lung expansion during inspiration.

Now take a deep breath and see if you can answer some questions:



EXAMPLE Q. The air that moves in and out of the lungs during normal, quiet breathing is called

- a. tidal volume.
- b. inspiratory reserve.
- c. vital capacity.
- d. lung capacity.
- e. residual volume.

A. The correct answer is tidal volume. The question asks only about air moved during normal, quiet breathing, not the kind of forceful air movement involved in measuring lung capacity. Think of the normal ebb and flow of the ocean's tide as opposed to the waves of a raging storm.

1 What does the mediastinum have to do with the lungs?

- a. It's the bone that protects them from collapsing during blunt trauma.
- b. It's the region between the right and left lungs.

- c. It's the control mechanism that reduces minimal air when the body is stressed.
- d. It's the membrane surrounding each lung.
- e. It's the muscle structure that moves the diaphragm to inflate and deflate them.

2 What is the purpose of mucus in the respiratory tract?

- a. to reduce friction of the air passing through
- b. to coat the alveoli to facilitate gas exchange
- c. to trap gas molecules for transport into the blood stream
- d. to trap dust and pollutants in the air for removal

3 Which term can be best described as the total amount of air than can be moved into or out of the lungs?

- a. tidal volume
- b. inspiratory reserve
- c. residual volume
- d. vital capacity
- e. total lung capacity

4 True or False: The lungs inflate with inspiration and deflate with expiration.

5 In which part of the respiratory system is cartilage never found?

- a. bronchioles
- b. primary bronchi
- c. secondary bronchi
- d. tertiary bronchi
- e. trachea

6-10 Fill in the blanks to complete the following sentences:

The trachea divides into two **6.**_____, which then divide into **7.**_____ with a branch going to each lobe of the lung. Upon entering the lobe, each divides into **8.**_____, subdividing into smaller tubes called **9.**_____. They terminate in an elongated sac called the atrium surrounded by **10.**_____ or air sacs.

11 If a pin were to pierce the body from the outside in the thoracic region, which three structures — in order — would it pass through?

- a. Lung, pleural cavity, terminal bronchioles
- b. Visceral pleura, lung, pleural cavity
- c. Parietal pleura, pleural cavity, visceral pleura
- d. Pleural cavity, trachea, lungs
- e. Primary bronchi, secondary bronchi, terminal bronchioles

12 Explain how normal inspiration is an active process while expiration is passive.

Preparing the Air

Before entering the trachea, air must first pass through the nose, pharynx, and larynx which help to prepare and filter the air. Though we are able to breathe through our mouths, that's more of a back-up plan so here we focus on the nose.

Knowing about the nose (and sinuses)

You may care a great deal about how your nose is shaped, but the shape actually makes little difference to your body. The nose is simply the most

visible part of your respiratory tract. Beyond those oh-so-familiar nostrils — which are formally called *external nares* — the *septum* divides the nasal cavity into two chambers. Inside the nostril is a slight dilation extending to the apex of the nose called the *vestibule*; it's lined by ciliated pseudostratified epithelium covered with hairs, plus mucous and sebaceous glands that help trap dust and particles before they can enter the lungs. The cilia sweep the mucus into the pharynx, where it is swallowed. Just under the lining are numerous blood vessels which serve to warm the air and the mucus moisturizes it.

Each nasal cavity is divided into an *olfactory region* and a *respiratory region*.

» **Olfactory region:** The olfactory region lies in the upper part of the nasal cavity. Fine filaments distributed over its mucous membrane are actually olfactory receptors devoted to the sense of smell. The bipolar olfactory cells' axons thread through openings in the *cribriform plate* (from the Latin *cribrum* for “sieve”) of the ethmoid bone and then come together in the *olfactory bulb*, which acts as an odorant classification filter (for perception of smells). These send signals to the higher olfactory centers of the brain's cerebral cortex for further discrimination and enhancement of the olfactory process. Impulses flow from the olfactory bulb through the olfactory tract to two main destinations:

- Via the *thalamus* to the *olfactory cortex* in the temporal lobe, where smells are consciously detected
- Via the sub-cortical route to the *hypothalamus* and to other regions of the limbic systems where emotional aspects of the smells are analyzed

» **Respiratory region:** The remainder of the nasal cavity comprises the respiratory region.



REMEMBER The respiratory region of the nasal cavity performs several important functions:

» It drains mucous secretions from the sinuses.

- » It drains lacrimal secretions from the eyes.
- » It prepares inhaled air for the lungs by warming, moistening, and filtering it. Dust and bacteria are caught in the mucus and passed outward from the nasal cavity by the motion of the cilia. Some of that gunk is taken up by *lymphatic tissue* in the nasal cavity and respiratory tubes for delivery to the lymph nodes, which destroy invading germs.

Ah, the sinuses. They can be such headaches. Lined with a pseudostratified ciliated columnar epithelium, *paranasal sinuses* are air-filled cavities in the bone that reduce the skull's weight and act as resonators for the voice. Each of the four groups of sinuses is named for the bone containing it (see [Chapter 7](#)), as follows:

- » **Frontal sinuses** are located in the frontal bone behind the eyebrows. (If you've ever flown with a sinus infection in an airplane, these are the suckers that hit you right behind the eyes.)
- » **Maxillary sinuses** are located in the *maxillae*, or upper jawbone.
- » **Ethmoid** and **sphenoid sinuses** are located in the ethmoid and sphenoid bones in the cranial cavity's floor.

Beyond the sinuses and connected to them are *nasal ducts* that extend from the medial angle of the eyes to the *nasal cavity*. These ducts let serous fluid from the eyes' *lacrimal glands* (which you know as tears) flow into the nasal cavity. Beyond the nasal cavity is the *nasopharynx*, which connects — you guessed it — the nasal cavity to the *pharynx*.

With a bit of a refresher on the nasal and sinus passages, do you think you can hit the following practice questions on the nose?

- 13** Which of the following statements about the mucous membranes of the nasal cavity is NOT true?
- a. They contain an abundant blood supply.
 - b. They moisten the air that flows over them.
 - c. They're composed of stratified squamous epithelium.
 - d. They filter the air of particulates.

14 Why does the nose have sebaceous glands?

- a. Their secretions smooth air flow into the nostrils.
- b. They enhance the sense of smell.
- c. Their secretions form a protective layer to warm, moisten, and help filter air.
- d. They improve resonance during speech.

15-29 Use the terms that follow to identify the structures of the respiratory tract shown in [Figure 13-1](#).

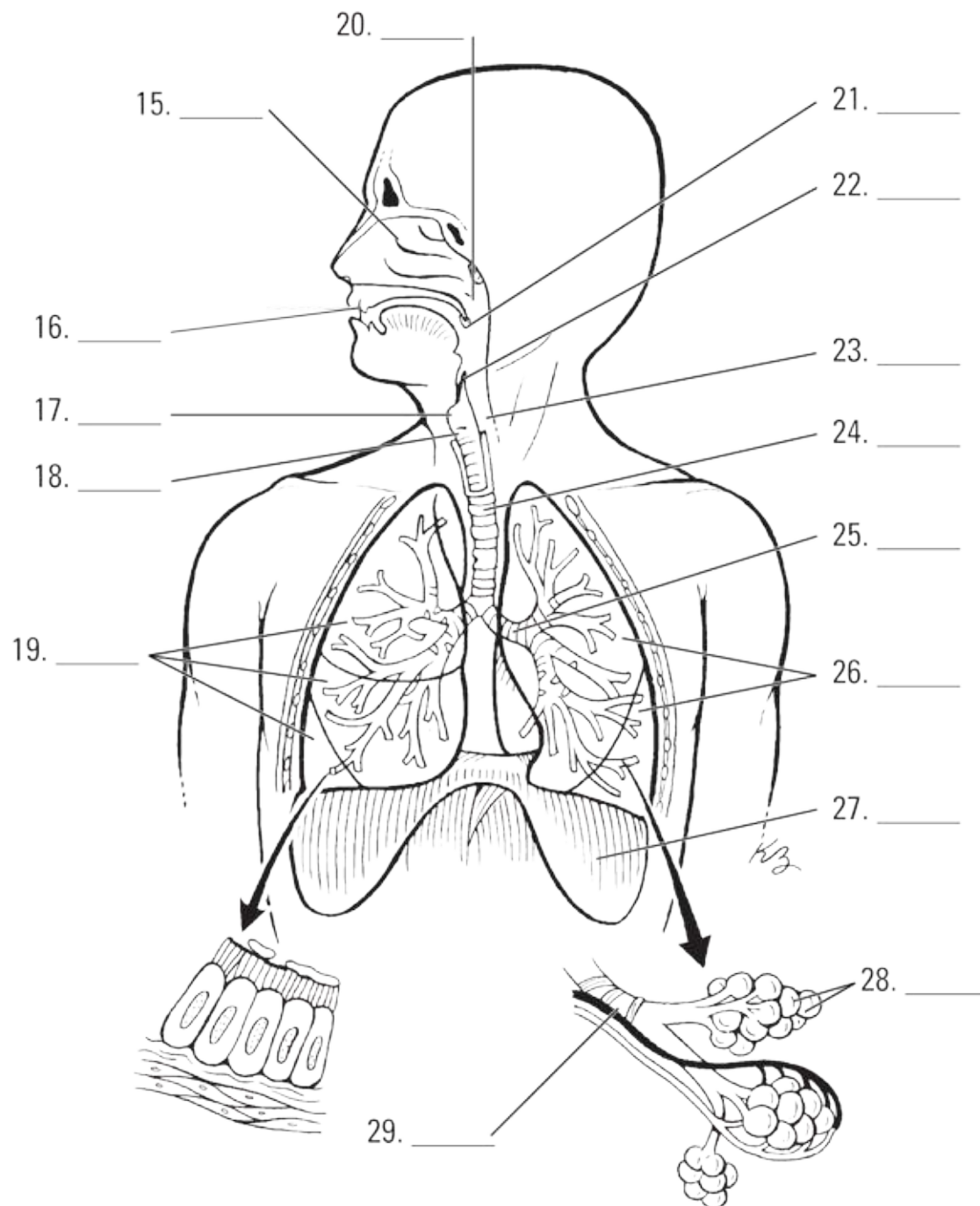


Illustration by Kathryn Born, MA

FIGURE 13-1: The respiratory tract.

- a. Esophagus
- b. Larynx
- c. Nasal cavity
- d. Oropharynx

- e. Diaphragm
- f. Right lung
- g. Epiglottis
- h. Mouth
- i. Alveoli
- j. Nasopharynx
- k. Thyroid cartilage
- l. Left lung
- m. Trachea
- n. Bronchioles
- o. Left primary bronchus

Dealing with throaty matters

In laymen's terms, it's the throat. But you know better, right? The "throat" consists of these key parts:

» **Pharynx:** The pharynx is an oval, fibromuscular tube about 5 inches long and tapering to $\frac{1}{2}$ inch in diameter at its *anteroposterior end*, which is a fancy biology term meaning "front to back." In fact, the point where the pharynx connects to the esophagus is the narrowest part of the entire digestive tract. The pharynx consists of three main divisions:

- The *nasopharynx*, where the nasal cavity opens into the pharynx
- The *oropharynx*, where the oral cavity opens into the pharynx
- The *laryngopharynx*, which begins at the epiglottis and leads down to the esophagus

Eustachian tubes connected to the middle ears enter the nasopharynx on each side. (And when the tubes become blocked by an infection, sudden pressure changes can be extremely painful.) On the back wall of the nasopharynx is a mass of lymphoid tissue called the *pharyngeal tonsil*, or *adenoids*.

» **Larynx:** Connecting the pharynx with the trachea, this collection of nine cartilages is what makes a man's prominent Adam's apple. Also called the

voice box, the larynx looks like a triangular box flattened dorsally and at the sides that becomes narrow and cylindrical toward the base. Ligaments connect the cartilages to several muscles, allowing their movements to be controlled. The inside of the larynx is lined with a mucous membrane that continues into the trachea.

Three of the larynx's nine cartilages go solo — the *thyroid*, the *cricoid*, and the *epiglottis* — while the others come in pairs — the *arytenoids*, the *corniculates*, and the *cuneiforms*.

- » **Thyroid cartilage:** The thyroid cartilage (*thyroid* in Greek means “shield-shaped”) is largest and consists of two plates called *laminae* that are fused just beneath the skin to form a shield-shaped process, the Adam's apple. Immediately above the Adam's apple, the laminae are separated by a V-shaped notch called the *superior thyroid notch*.
- » **Cricoid cartilage:** The ring-shaped cricoid cartilage is smaller but thicker and stronger, with shallow notches at the top of its broad back that connect, or articulate, with the base of the arytenoid cartilages.
- » **Arytenoid cartilages:** The arytenoid cartilages are both shaped like pyramids, with the vocal folds attached at the back and the controlling arytenoid muscles that move the arytenoids attached at the sides, moving the vocal cords.
- » **Corniculate cartilages:** On top of the arytenoids are the corniculate cartilages, small conical structures for attachment of muscles regulating the tension of the vocal cords.
- » **Cuneiform cartilages:** Nestled in front of these and inside the *aryepiglottic* fold, the cuneiform cartilages stiffen the soft tissues in the vicinity.
- » **Epiglottis cartilage:** The epiglottis, sometimes called the lid on the voice box, is formed from elastic cartilage in the shape of a leaf. It is attached at its stem end and projects upward behind the root of the tongue. It opens during respiration and reflexively closes during swallowing to keep food and liquids from getting into the respiratory tract. Its elastic cartilage gives it a flexible attachment to the anterior and superior borders of the *hyoid bone* and larynx.

Two types of folds play different roles inside the larynx.

- » The *true vocal folds*, or cords, are V-shaped when relaxed. When talking, the folds stretch for high sounds or slacken for low sounds, causing the glottis — the opening in the larynx — to form an oval. Sounds are produced when air is forced over the folds, causing them to vibrate.
- » Just above these folds are the *vestibular folds*, also known as the false vocal cords, that don't produce sounds. Muscle fibers within these folds help close the glottis during swallowing — the last chance for preventing food from entering the trachea.

Following are some practice questions dealing with the throat:

30 What happens when the arytenoids move?

- a. The Adam's apple moves.
- b. The cricoid cartilage flexes.
- c. The vocal cords move.
- d. The epiglottis closes.
- e. The cuneiform cartilages stiffen.

31-35 Match the anatomical structure with its function.

- a. Voice box lid
- b. Respiratory center
- c. Prevent collapse of trachea
- d. Fight germs
- e. Adam's apple

- 31. _____ Epiglottis
- 32. _____ Adenoids
- 33. _____ Medulla oblongata
- 34. _____ C-shaped cartilaginous rings
- 35. _____ Thyroid cartilage

36 The _____ is the opening between the two vocal folds, and the _____ covers that opening as needed.

- a. epiglottis, glottis
- b. bronchi, epiglottis
- c. alveoli, pharynx
- d. glottis, pharynx
- e. glottis, epiglottis

37-49 Use the terms that follow to identify the structures of the larynx shown in [Figure 13-2](#). **Note:** Some terms may be used more than once.

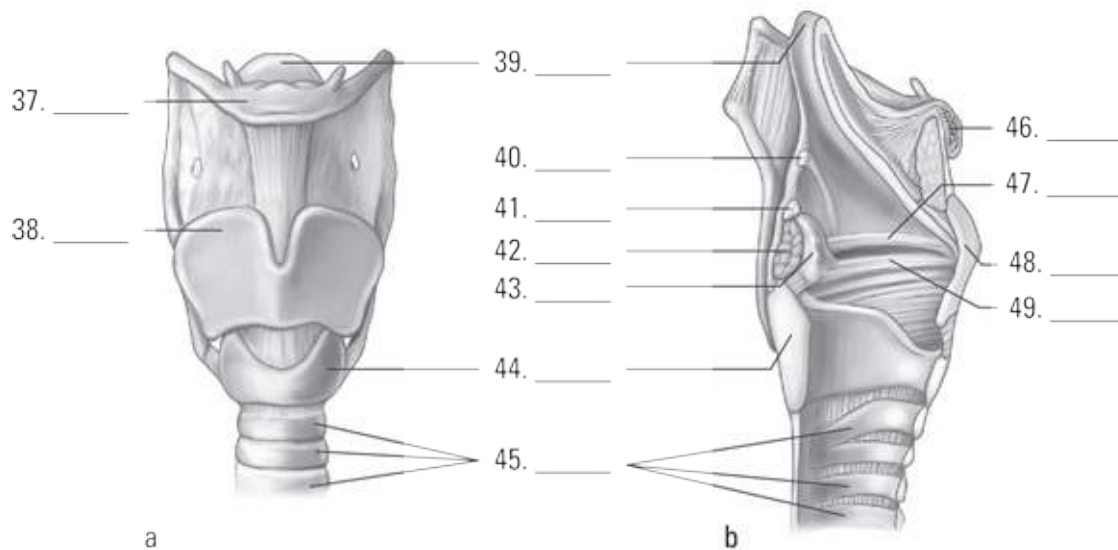


Illustration by Imagineering Media Services Inc.

FIGURE 13-2: Front (a) and lateral (b) views of the larynx.

- a. Thyroid cartilage
- b. Cricoid cartilage
- c. Hyoid bone
- d. Epiglottis
- e. Arytenoid cartilage
- f. Vestibular fold (false vocal cord)
- g. Corniculate cartilage

- h. Cuneiform cartilage
- i. Arytenoid muscle
- j. True vocal cord
- k. C-shaped tracheal cartilages

Swapping the bad for the good

Now that we know how the air is prepared, and enters and leaves the lungs, it's time to look at what happens while it's there. The final destination of the air is the *alveoli* (singular: *alveolus*). At the end of the terminal bronchioles are the *alveolar ducts* which route air into the *alveolar sac*. Linked to this are the alveoli forming what looks like a cluster of grapes — with each alveolus being a single grape. Wrapped around each cluster of alveoli is a network of capillaries — this is the site of gas exchange which you can see illustrated in [Figure 13-3](#). The numerous alveoli create a large surface area for this.

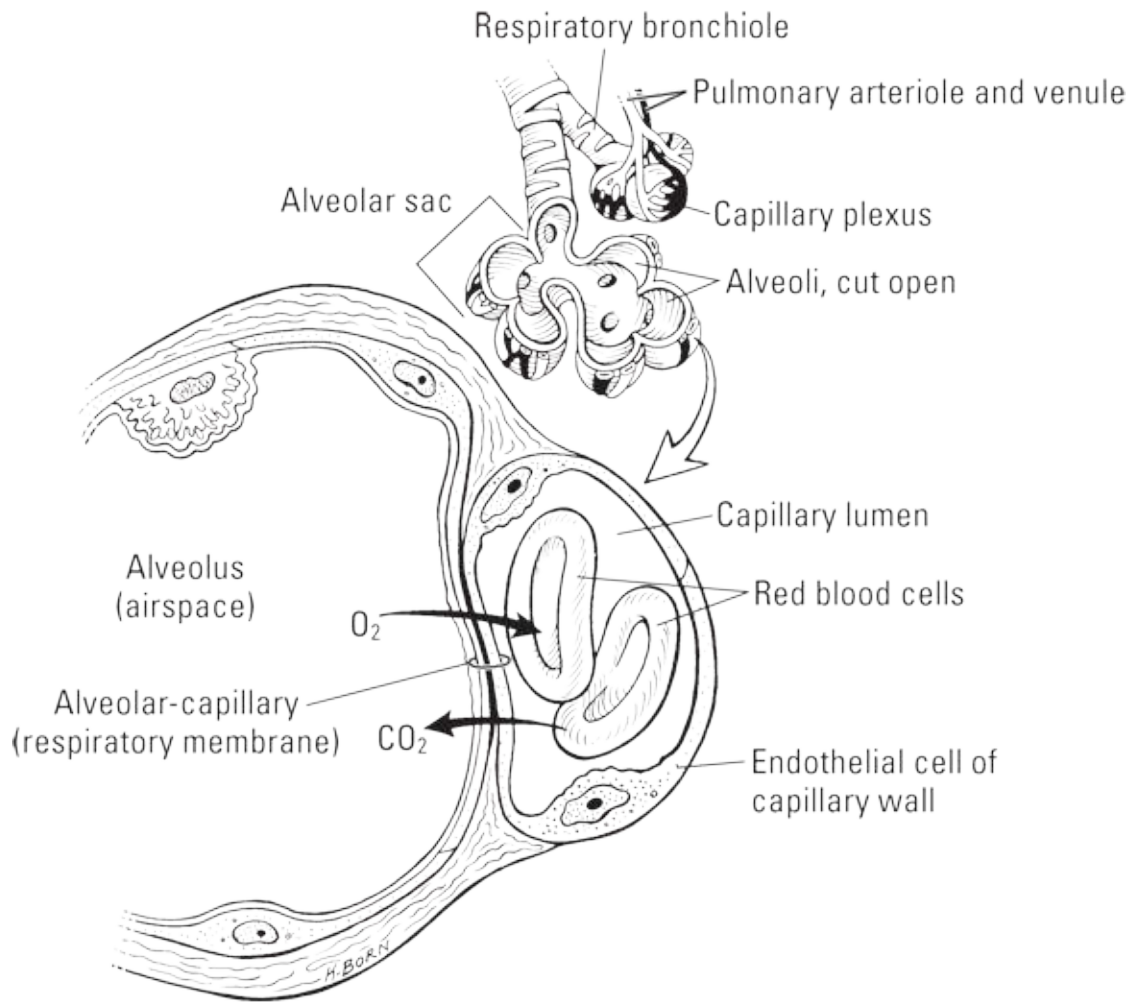


Illustration by Kathryn Born, MA

FIGURE 13-3: Oxygen–carbon dioxide exchange in the lung.

Like the walls of the capillaries (see [Chapter 11](#)), the walls of the alveoli are simple squamous. This means the *respiratory membrane* is merely two layers of thin, flattened cells thick. With only two layers of cells between the air and the blood, gas exchange occurs easily due to the differences in partial pressure of O_2 and CO_2 (which you can think of as a concentration gradient). Because the partial pressure of CO_2 (pCO_2) is lower in the air than in the blood, the molecules diffuse through the respiratory membrane into the air in the alveoli to be exhaled. Since the partial pressure of O_2 (pO_2) is higher in the air, it diffuses readily into the blood where it is picked up by hemoglobin in the red blood cells to be carried to the tissues.



REMEMBER Blood comes to the lungs through two sources: the *pulmonary arteries* and the *bronchial arteries*. The pulmonary trunk comes from the right ventricle of the heart and then branches into the two pulmonary arteries carrying deoxygenated blood (the only arteries that contain blood loaded with carbon dioxide) from various parts of the body to the lungs (see [Chapter 11](#) for an introduction to the cardiovascular system). That blood goes through pulmonary arterioles and then through pulmonary capillaries for gas exchange. After that, oxygenated blood leaves the lung through pulmonary venules, returning to the left atrium through the *pulmonary veins* (the only veins that contain oxygenated blood), completing the cycle. Bronchial arteries branch off the thoracic aorta of the heart, supplying the lung tissue with nutrients and oxygen.

See whether you're carrying away enough information about respiration by tackling the following questions:

50 What is hemoglobin's primary role?

- a. Filtration
- b. Gaseous transport
- c. Ventilation
- d. Asphyxia

51-54 Fill in the blanks to complete the following sentences:

Upon inhalation, molecules of **51.**_____ diffuse into the lung's tissues. From there, these molecules then diffuse into **52.**_____ cells, which contain a pigment called **53.**_____. Simultaneously, a second substance formed during cellular respiration, **54.**_____, is released into the lungs to be expelled during exhalation.

55-61 Use the terms that follow to identify the structures of the bronchiole shown in [Figure 13-4](#).

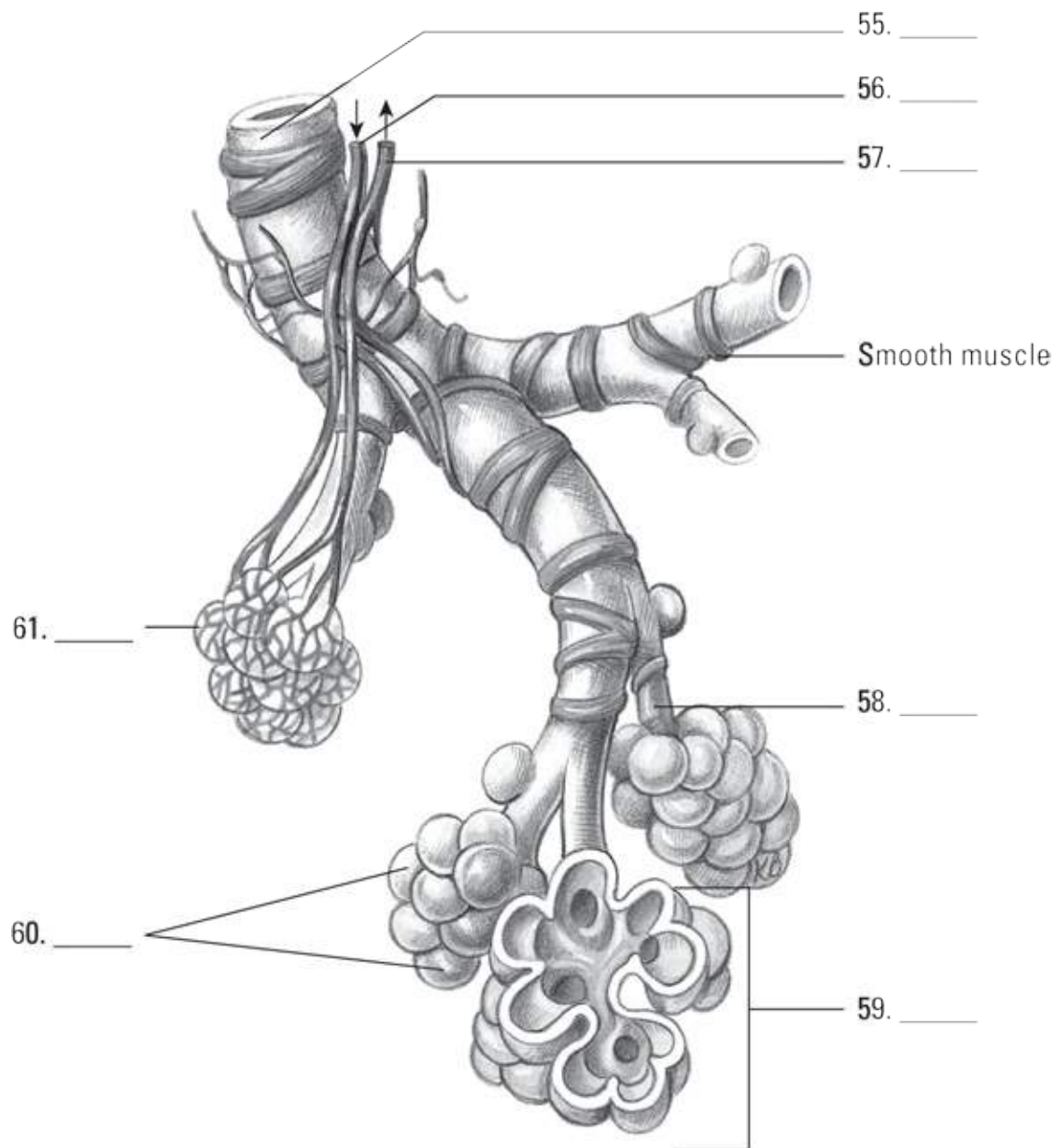


Illustration by Kathryn Born, MA

FIGURE 13-4: A bronchiole.

- a. Pulmonary venule
- b. Alveolar sac
- c. Tertiary bronchiole
- d. Pulmonary capillary
- e. Pulmonary arteriole
- f. Alveolar duct

g. Alveoli

- 62** Blood in the pulmonary artery has a pO_2 of 40mmHg and a pCO_2 of 46mmHg. The air in the alveoli has a pO_2 of 105mmHg and a pCO_2 of 40mmHg. Which statement best describes what will occur?
- a. Both O_2 and CO_2 will exit the alveoli.
 - b. Both O_2 and CO_2 will enter the alveoli.
 - c. O_2 will enter the alveoli and CO_2 will enter the blood.
 - d. O_2 will exit the alveoli and CO_2 will exit the blood.

Answers to Questions on the Respiratory System

The following are answers to the practice questions presented in this chapter.

- ① What does the mediastinum have to do with the lungs? **b. It's the region between the right and left lungs.** It encases everything contained in the thoracic cavity except for the lungs.
- ② What is the purpose of mucus in the respiratory tract? **d. to trap dust and pollutants in the air for removal**
- ③ Which term can be best described as the total amount of air than can be moved into or out of the lungs? **d. vital capacity.** The total lung capacity includes the residual volume which cannot ever exit the lungs.
- ④ True or False: The lungs inflate with inspiration and deflate with expiration. **False.** If all the air exited a lung, it would collapse — and so would the alveoli, meaning no gas exchange can occur.
- ⑤ In which part of the respiratory system is cartilage never found? **a. Bronchioles.** They're so small that they need to be more elastic than cartilage would allow them to be.
- ⑥-10 The trachea divides into two **6. primary bronchi**, which then divide into

7. **secondary bronchi** with a branch going to each lobe of the lung. Upon entering the lobe, each divides into 8. **tertiary bronchi**, subdividing into smaller tubes called 9. **bronchioles**. They terminate in an elongated sac called the atrium surrounded by 10. **alveoli**, or air sacs.

- 11) If a pin were to pierce the body from the outside in the thoracic region, which three structures — in order — would it pass through? **c. Parietal pleura, pleural cavity, visceral pleura**. Note that the question asks you to choose from the lists provided, not from the entire structure of the body.
- 12) Explain how normal inspiration is an active process while expiration is passive. **Breathing in requires muscle contraction which requires energy, hence active. Even without conscious control, the diaphragm contracts decreasing pressure inside of the lungs. From here, no more energy is expended. Air flows into the lungs because of the pressure differential. The medulla oblongata stops stimulating the diaphragm so it relaxes. That paired with the elastic recoil of the lung tissue increases pressure so air flows out; no contraction, no energy.**
- 13) Which of the following statements about the mucous membranes of the nasal cavity is not true? **c. They're composed of stratified squamous epithelium**. They are ciliated pseudostratified like most linings that secrete mucus from goblet cells.
- 14) Why does the nose have sebaceous glands? **c. Their secretions form a protective layer to warm, moisten, and help filter air.**
- 15-29) Following is how [Figure 13-1](#), the respiratory tract, should be labeled.
15. **c. Nasal cavity**; 16. **h. Mouth**; 17. **k. Thyroid cartilage**; 18. **b. Larynx**; 19 **f. Right lung**; 20. **j. Nasopharynx**; 21. **d. Oropharynx**; 22. **g. Epiglottis**; 23. **a. Esophagus**; 24. **m. Trachea**; 25. **o. Left primary bronchus**; 26. **l. Left lung**; 27. **e. Diaphragm**; 28. **i. Alveoli**; 29 **n. Bronchioles**
- 30) What happens when the arytenoids move? **c. The vocal cords move**. This cartilage is tough to spell and pronounce but easy to move.
- 31) Epiglottis: **a. Voice box lid**

- 32) Adenoids: **d. Fight germs**
- 33) Medulla oblongata: **b. Respiratory center**
- 34) C-shaped cartilaginous rings: **c. Prevent collapse of trachea**
- 35) Thyroid cartilage: **e. Adam's apple**
- 36) The **e. glottis** is the opening between the two vocal folds and the **e. epiglottis** covers that opening as needed. **Note:** The prefix *epi* means “on, upon, or above.”

37-49) Following is how [Figure 13-2](#), the larynx, should be labeled.

37. **c. Hyoid bone**; 38. **a. Thyroid cartilage**; 39. **d. Epiglottis**; 40. **h. Cuneiform cartilage**; 41. **g. Corniculate cartilage**; 42. **e. Arytenoid cartilage**; 43. **i. Arytenoid muscle**; 44. **b. Cricoid cartilage**; 45. **k. C-shaped tracheal cartilages**; 46. **c. Hyoid bone**; 47. **f. Vestibular fold (false vocal cord)**; 48. **a. Thyroid cartilage**; 49. **j. True vocal cord**

- 50) What is hemoglobin's primary role? **b. Gaseous transport.** Hemoglobin is the protein that bonds with and releases gasses within the bloodstream. That's how oxygen gets where it needs to go and how carbon dioxide gets shipped off to the lungs for waste removal.



TIP Remember that the Latin root for blood is *hemo*; other than filtration, which occurs elsewhere, none of the other answer options apply to blood.

- 51-54) Upon inhalation, molecules of **51. oxygen** diffuse into the lung's tissues. From there, these molecules then diffuse into **52. red blood** cells, which contain a pigment called **53. hemoglobin**. Simultaneously, a second substance formed during cellular respiration, **54. carbon dioxide**, is released into the lungs to be expelled during exhalation.

55-61) Following is how [Figure 13-4](#), the bronchiole, should be labeled.

55. **c. Tertiary bronchiole**; 56. **e. Pulmonary arteriole**; 57. **a. Pulmonary venule**; 58. **f. Alveolar duct**; 59. **b. Alveolar sac**; 60. **g. Alveoli**; 61. **d.**

Pulmonary capillary

- 62) Blood in the pulmonary artery has a pO_2 of 40mmHg and a pCO_2 of 46mmHg. The air in the alveoli has a pO_2 of 105mmHg and a pCO_2 of 40mmHg. Which statement best describes what will occur? **d. O_2 will exit the alveoli and CO_2 will exit the blood.** Gasses will always travel to where there is a lower partial pressure (which is just like the concentration gradient) if they're able to move through whatever barrier is in the way. Since the respiratory membrane is only two cells thick, oxygen and carbon dioxide diffuse through it easily, all without the use of ATP.